

Nonreciprocal Breathing Solitons

Jonas Veenstra^{1,*} Oleksandr Gamayun^{2,*} Martin Brandenbourger^{1,3,*} Freek van Gorp¹
 Hans Terwisscha-Dekker,¹ Jean-Sébastien Caux¹ and Corentin Coulais¹

¹*Institute of Physics, Universiteit van Amsterdam,
 Science Park 904, 1098 XH Amsterdam, The Netherlands*

²*London Institute for Mathematical Sciences, Royal Institution,
 21 Albemarle Street, London W1S 4BS, United Kingdom*

³*Aix-Marseille Université, CNRS, Centrale Méditerranée, IRPHE, UMR 7342, 13384 Marseille, France*



(Received 17 May 2024; accepted 10 July 2025; published 18 August 2025)

Breathing solitons consist of a fast beating wave within a compact envelope of stable shape and velocity. They can propagate and carry information and energy in a variety of contexts such as plasmas, optical fibers, and cold atoms, but propagating breathers have remained elusive when energy conservation is broken. Here, we report on the observation of breathing, unidirectional, arbitrarily long-lived solitons in nonreciprocal, nonconservative active metamaterials. Combining precision desktop experiments, numerical simulations, and perturbation theory on generalizations of the sine-Gordon and nonlinear Schrödinger equations, we demonstrate that unidirectional breathers generically emerge in weakly nonlinear nonreciprocal materials, and that their dynamics are governed by an unstable fixed point. Crucially, breathing solitons can persist for arbitrarily long times provided that (i) this fixed point displays a bifurcation when a delicate balance between energy injection and dissipation is struck and (ii) the initial conditions allow the dynamics to reach this bifurcation point. Importantly, discrete effects tend to stabilize these nonreciprocal breathers over a wider range of initial conditions. Our work establishes nonreciprocity as a promising avenue to generate stable nonlinear unidirectional waves and could be generalized beyond metamaterials to optics, soft matter, and superconducting circuits.

DOI: [10.1103/nrv2-9h8z](https://doi.org/10.1103/nrv2-9h8z)

Subject Areas: Metamaterials, Nonlinear Dynamics,
 Soft Matter

I. INTRODUCTION

Breaking the symmetry of interactions is commonplace in complex systems as diverse as prey-predator interactions [1,2], networks of neurons [3], and human behavior [4,5]. This idea has recently seen an explosion of activity across physics. Nonreciprocity has been theoretically introduced in synthetic many-body systems such as quantum [6–11] and spin systems [12], and experimentally realized in optical devices [13–16], plasmonic nanoparticles under optical drive [17], colloids [18,19], emulsions [20–22], and metamaterials [23–29]. In these systems, nonreciprocity is typically achieved by injecting momentum or energy in a way that breaks the symmetry or hermiticity of the linear operators governing the dynamics [30–32] and entails that the transfer function differs if you swap input and output.

Such asymmetry then often leads to unidirectional or chiral amplification, necessarily pushing the dynamics beyond the linear regime into one where nonlinearities dominate and features like solitons can emerge. Nonlinear nonreciprocal dynamics have recently been investigated in the context of pattern formation [33–35], topological excitations and defects [18,19,28,36], shocks [21], turbulence [37], and topological edge modes [38,39]. Yet, nonreciprocal dynamics of localized nontopological nonlinear excitations such as solitons have remained largely unexplored.

Nontopological solitons are stable and localized weakly nonlinear excitations [40] and can exist in media that are driven far from equilibrium [41], in which case they are often called dissipative solitons. Common examples of such driven-dissipative media include plasmas [42], optical microcavities [43–45], granular materials [46–48], and mechanical metamaterials [28,49–61]. In contrast to solitons in conservative systems, dissipative solitons must strike a balance between loss and gain to be stable. In contrast to topological dissipative solitons, nontopological solitons must strike an additional balance between dispersion and nonlinearity. These two balances put severe

*These authors contributed equally to this work.

Published by the American Physical Society under the terms of the [Creative Commons Attribution 4.0 International](https://creativecommons.org/licenses/by/4.0/) license. Further distribution of this work must maintain attribution to the author(s) and the published article's title, journal citation, and DOI.

constraints on the lifetime and dynamics of dissipative solitons [41,62]. A natural set of questions hence arises: What types of solitons emerge in weakly nonlinear nonreciprocal materials? Which mechanisms govern their dynamics?

Here, we address these questions by focusing on breathing solitons. Breathing solitons are localized nonlinear excitations, modulated by a carrier wave. They can either be standing—in which case they are sometimes called oscillons—or traveling—in which case they can be used to transport information and energy. Standing breathing solitons have been observed in a plethora of dissipative systems, from optical cavities [43,44,63] and arrays of nanoplasmonic particles in the presence of driving and dissipation [64–67] to driven granular media [68] and fluids [69]. In stark contrast, traveling breathing solitons have yet to be observed in strongly dissipative systems. The only exception can be found in nanoplasmonic particle arrays, where traveling patterns reminiscent of breathing solitons have been observed numerically under a unidirectional external drive [65], yet without experiments or theoretical description of the solitons.

We discover here that traveling breathing solitons can spontaneously emerge in nonreciprocal media. Using experiments on active metamaterials, numerical simulations, and perturbation theory, we find that a delicate balance between initial conditions, nonreciprocal gain, and damping sets the conditions for the emergence and lifetime of these solitons. This balance determines whether radiation, one breather, multiple breathers, or kink-antikink pairs are formed following an initial perturbation; together with the magnitude and velocity of the initial condition, this balance also determines the fate of the breathing soliton.

Strikingly, we find that in contrast to breather solitons in the presence of constant driving [43,44,63] (see Appendix C 5), nonreciprocity makes the dissipative breathers travel and expands the regime of existence of traditional breathing dissipative solitons. Our results establish active metamaterials as a productive avenue for the study of unidirectional dissipative solitons and for nonlinear generalizations of nonreciprocal excitations.

II. BREATHERLIKE WAVE IN A NONRECIPROCAL ACTIVE METAMATERIAL

Consider the active mechanical metamaterial shown in Fig. 1(a) and in Supplemental Material Video 1 [70]. It is made of 50 oscillators, each linked to its neighbors by a thin flexible rubber plate (see Appendix A for details). Upon relative rotation of neighboring oscillators θ_i and θ_{i+1} , this plate will bend and stretch and thereby apply a nonlinear elastic restoring torque on oscillator i , $\tau_i^e(\theta_i, \theta_{i+1})$. This restoring torque is symmetrical by virtue of Newton's third law $\tau_i^e(\theta_i, \theta_{i+1}) = -\tau_{i+1}^e(\theta_i, \theta_{i+1})$. By design, we break Newton's third law by applying an external torque on each oscillator. This torque depends on the configuration of the

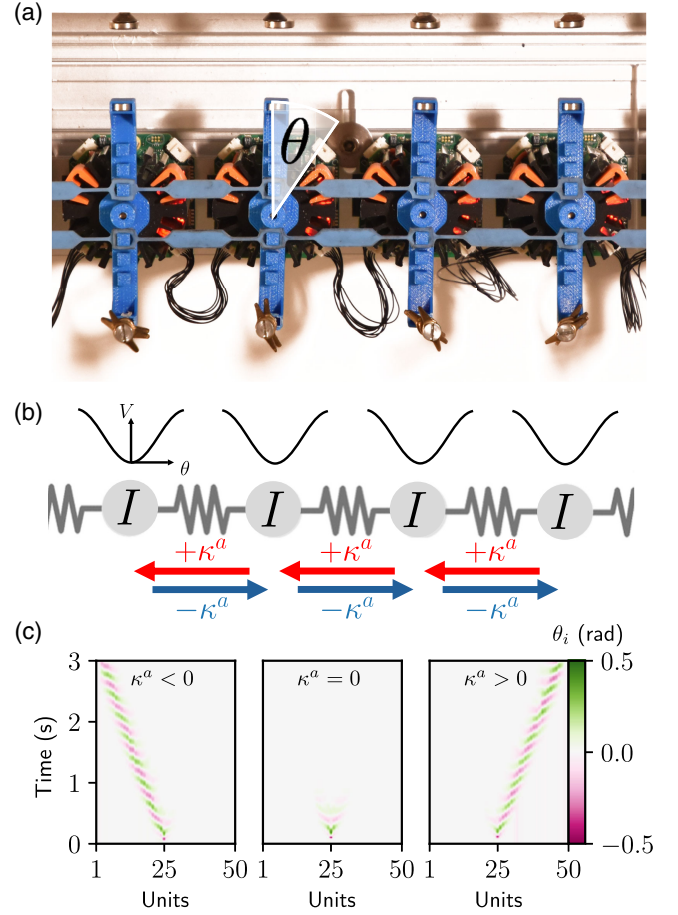


FIG. 1. Breathing nonreciprocal soliton. (a) Picture of four identical unit cells composing the active mechanical metamaterial. Each unit has one rotational degree of freedom (vertical blue bar) which is elastically coupled to its nearest neighbors via the horizontal thin blue elastic bands. The nonreciprocal gain is programmed by the microcontroller embedded in each unit (Appendix A). Magnets placed on the oscillator tips and a periodic substrate generate a weakly nonlinear potential. The moment of inertia is controlled by weights to the oscillator tips (golden bolts). (b) Schematic overview of the metamaterial whose rotational degrees of freedom have moment of inertia I , are connected by springs offset by nonreciprocity κ^a (red and blue arrows), and are each subject to a weakly nonlinear magnetic potential V . (c) Kymograph of the angular deviations θ_i of 50 connected oscillators upon excitation of the middle oscillator. Without nonreciprocal gain κ^a , the excitation quickly disperses and dies out (middle panel). For positive (negative) gain, a breathing localized wave forms and travels to the right (left), as seen in the right-hand (left-hand) panel.

system $\tau_i^a = \kappa^a(\theta_{i-1} - \theta_{i+1})$, where increasing i corresponds to the clockwise direction. This is done by using an active control system that relies on motors, sensors, and microcontrollers embedded in our oscillators. This active torque introduces a nonreciprocal bias in the system and amplifies waves and pulses unidirectionally, in what is known as the non-Hermitian skin effect [23,30–32]. As unidirectional amplification occurs, the magnitude of the

signal may increase and nonlinearities will inevitably come into play. In order to probe the nonlinear response in a controlled way, we place magnets both on the oscillator tips and periodically on a substrate [28], which creates a weakly nonlinear potential [Fig. 1(b) and see Appendix A]. We first probe the response of the active mechanical metamaterial to an initial perturbation which we apply manually (see Supplemental Material Video 1 [70]). Without nonreciprocity $\kappa^a = 0$, the perturbation disperses symmetrically around the locus of impact and dissipates within a fraction of a second [Fig. 1(c)]. For nonzero κ^a , we observe that the perturbation transforms into a localized and oscillating wave packet that propagates at a constant velocity of 8 ± 1 units/s in a direction given by the sign of κ^a . Strikingly, nonreciprocity mitigates the dispersive and dissipative effects and gives rise to a traveling and localized oscillating wave whose lifetime can be extended to multiple seconds depending sensitively on the strength of κ^a . Besides the stable envelope, this wave has a carrier wave that is oscillating at a frequency of 5 Hz and moving at a constant phase velocity of 28 units/s.

Under which conditions can nonreciprocity sustain this nonlinear excitation? To ascertain this, we perform a set of experiments in which we systematically vary the non-dimensional gain $\eta \propto \kappa^a/\kappa$ and the perturbation velocity $\dot{\theta}_{t=0}$ [Figs. 2(a) and 2(b)], now controlled by a pulse exerted by the unit's internal motor. This reveals the strong influence of the nonreciprocal gain: When the gain is too small, the perturbation disperses asymmetrically but quickly dissipates; when η is too large, oscillators are rendered unstable and a rapidly amplifying wave emerges.

Only when the nonreciprocity is carefully tuned can a localized wave of constant amplitude be observed that lasts several seconds.

Why do these waves emerge from nonreciprocal interactions? And what is their nature? To answer these questions, we first consolidate our experimental findings using a discrete model that captures all the essential features of the nonreciprocal active material. We model the dynamics of the interacting oscillators subject to a nonlinear potential by a nonreciprocal version of the Frenkel-Kontorova model, which reads in nondimensional form (see Appendix B)

$$\ddot{\phi}_i = \phi_{i-1} + \phi_{i+1} - 2\phi_i - \frac{\eta}{2}(\phi_{i+1} - \phi_{i-1}) - \Gamma\dot{\phi}_i - D \sin(\phi_i), \quad (1)$$

where $\phi_i = 2\pi(\theta_i/\theta_d)$ denotes the i th oscillator angle normalized by the period of the magnetic potential $\theta_d = 1$ rad. The nondimensional parameters $\Gamma = 0.3 \pm 0.1$ and $D = 1.9 \pm 0.3$ represent on-site dissipation of the oscillators and magnetic potential strength and $\eta \propto \kappa^a/\kappa$ whose values we calibrate experimentally using a torsion testing machine (see Appendix A for details). We numerically integrate this equation and find a quantitative

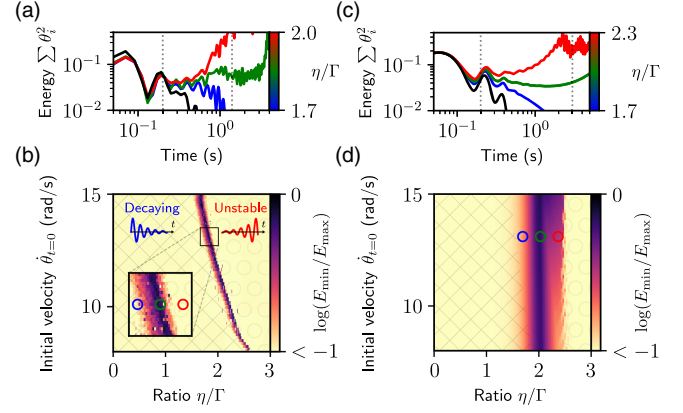


FIG. 2. Three regimes of breathers. (a) The total squared amplitude of the oscillators over time for a range of the nondimensional gain η normalized by the viscous dissipation $\Gamma = 0.3 \pm 0.1$ shows decaying (blue), diverging (red), and long-lived (green) breathers. Black line shows the passive case $\eta = 0$. As a metric for the longevity of the breather, we use the ratio $E_{\min}/E_{\max}$ between minimum and maximum energy measured within a window demarcated by the dotted lines. (b) A sweep over initial conditions $\dot{\theta}_{t=0}$ and η/Γ shows a narrow regime of extended breather lifetime, quantified by the logarithm of the ratio between the minimum and maximum amplitudes of the breather measured within a time period of 1 s, indicated by the dashed vertical lines in (a) (see Appendix A). Square and circularly hatched areas demarcate regions of stable decaying and unstable amplifying waves, respectively. Inset shows a zoom-in of the extended lifetime regime. (c) Simulations of Eq. (1) using experimental values $\Gamma = 0.3$ and $D = 1.9$. (d) A sweep over the same initial conditions reveals a narrow range of breathers with an extended lifetime.

agreement with the experiments [Figs. 2(b)–2(d)] with additional insights from the greater precision of our numerical simulations. First, the velocity of the breathing waves does not sensitively depend on the level of nonreciprocity [Fig. 2(b)]. Second, long-lasting breathing waves exist only for a narrow regime of nonreciprocity, and in this regime they emerge for a wide range of perturbation velocities. Higher velocity perturbations extend the region of stable solutions to a wider range of the nonreciprocal gain. This suggests that the existence of long-lasting unidirectional nonlinear waves depends on a precise balance between gain and loss.

III. NONRECIPROCAL BREATHING SOLITONS

To confirm that a nonlinear nonreciprocal material can host breathing solitons, we take the continuum limit of Eq. (1) and find a sine-Gordon equation (see Appendix C 1), augmented with nonreciprocity and damping:

$$\varphi_{tt} - \varphi_{xx} + \sin(\varphi) = -\eta\varphi_x - \Gamma\varphi_t. \quad (2)$$

While a related version of this equation with a constant driving instead of a nonreciprocal term has been studied in

detail [40,71], breathers in its nonreciprocal counterpart have to the best of our knowledge remained unexplored. Crucially, the left-hand side of Eq. (2) is integrable, which will allow us to unambiguously identify the breathers (as well as the other discrete structures) and explore their full evolution in Eq. (2) by means of the perturbation theory applied to the auxiliary linear scattering problem.

We perform a numerical investigation of profile dynamics governed by Eq. (2) for a fixed value of the damping $\Gamma = 0.1$ and various parameters of the nonreciprocity η . We use periodic boundary conditions, in accordance with the experiments and numerical simulations above. To simulate the “kick” initial condition, we impose

$$\varphi(x, 0) = 0, \quad \partial_t \varphi(x, 0) = \frac{\Pi}{\cosh(x/d)}. \quad (3)$$

We have also tried other forms of kicks including a step function and a Gaussian profile and found that breathers consistently emerge provided a balance between nonreciprocity and damping exists, and provided a sufficiently large kick is given. For our chosen profile Eq. (3), the injected energy is controlled by the magnitude Π of the applied kick and its width d which is set to 1 in all simulations.

Similarly to the discrete case and to the experiment, we find qualitatively the same breatherlike behavior [Fig. 3(a)]: a travelling pulse with a steady envelope and a rapidly oscillating carrier wave. We also observe the three regimes identified earlier, where for a relatively small nonreciprocity to dissipation ratio $\eta/\Gamma = 1.00$, the wave decays; when this ratio becomes larger $\eta/\Gamma = 1.50$, the wave blows up; and in between at $\eta/\Gamma = 1.37$, the wave survives for a remarkably long time [Fig. 3(b)]. In agreement with experiment, we find that the velocity of the wave does not depend on nonreciprocity [Fig. 3(b)], contrary to the case of topological solitons [28].

A systematic investigation of the instantaneous energy of the profile,

$$\mathcal{E} = \frac{1}{2} \int \left(\frac{(\partial_t \varphi)^2 + (\partial_x \varphi)^2}{2} + 1 - \cos \varphi \right) dx, \quad (4)$$

at every moment of time allows us to compute the maximum variation in energy of the nonlinear wave over the course of time, and reveals that there are two regions where the nonlinear wave is particularly long-lived: (i) a specific region $\eta/\Gamma \simeq 1.4$ and $\Pi \simeq 4.2$ [inset of Fig. 3(c)] and (ii) a large region $\Pi > 4.4$ and $\eta < 1.4$.

But are these nonlinear waves breathing solitons? To address this question, we perform a spectral decomposition according to the left-hand side of Eq. (2). Indeed, the unperturbed part is an integrable equation that can be solved exactly by the technique based on the inverse scattering transform (IST) [72,73]. This procedure can be understood as a nonlinear Fourier transformation that

transforms the profile $\varphi(x)$ into the scattering data of the auxiliary linear problem with the potential that depends on $\varphi(x)$ (so-called direct scattering). It is convenient instead of the transmission and reflection coefficients to present the data in the functions $a(\lambda)$ and $b(\lambda)$ of the spectral parameter λ . The time evolution of these variables is extremely simple (see Appendix C 2 for details).

The breather appears whenever $a(\lambda)$ has zeros in the upper half-plane (we label these points $\lambda_1 = \zeta + i\nu$, $\nu > 0$, and $\lambda_2 = -\bar{\lambda}_1$). The corresponding profile reads

$$\varphi(x, t) = -4 \arctan \frac{\lambda_1 - \bar{\lambda}_1}{\lambda_1 + \bar{\lambda}_1} \frac{\mathbf{g}(x, t) - \bar{\mathbf{g}}(x, t)}{1 + |\mathbf{g}(x, t)|^2}, \quad (5)$$

with

$$\mathbf{g}(x, t) = \exp \left(\frac{ix(\lambda_1^2 - 1)}{2\lambda_1} + \frac{it(\lambda_1^2 + 1)}{2\lambda_1} \right) \mathbf{g}(0). \quad (6)$$

If the right-hand side of Eq. (2) were zero, the zeros λ_1 and λ_2 would be constant complex numbers. In the present case, in the adiabatic approximation [72], the right-hand side leads to modification of the parameters of the breather; i.e., λ_1 and λ_2 start depending on time. This means that the features of the solitons such as amplitude and velocity evolve over time as a consequence of dissipation and nonreciprocity.

The scattering data of the initial profile Eq. (3) can be found explicitly [see Eqs. (C29) and (C30) of Appendix C]. The breather is present in the initial profile if $2 < \Pi d < 4$ and the corresponding zeros are given by

$$\lambda_{1,2} = i \frac{\Pi d - 2}{2d} \pm \sqrt{1 - \left(\frac{\Pi d - 2}{2d} \right)^2}. \quad (7)$$

In the unperturbed system, these spectral data would correspond to the profile

$$\varphi = 4 \arctan \frac{\frac{\Pi d - 2}{2d} \sin t \sqrt{1 - \left(\frac{\Pi d - 2}{2d} \right)^2}}{\sqrt{1 - \left(\frac{\Pi d - 2}{2d} \right)^2} \cosh \frac{\Pi d - 2}{2d} x}. \quad (8)$$

For the time-evolved profile the corresponding scattering data can be computed numerically. We employ the transfer-matrix method described in Ref. [74]. Strictly speaking the initial condition for the profile corresponds to a kink-antikink pair—where λ_1 and λ_2 are purely imaginary, rather than a breather—where $i\lambda_1$ and $i\lambda_2$ are conjugate complex numbers. In the simulation, however, this pair immediately transforms into a breather because of the dissipative terms, which is reflected in the nonzero real part in Fig. 3(d). Note also that for large nonreciprocity the kink-antikink pair profile might persist for short times (red curve until $t = 3$) and return at later times (red curve at time $t \simeq 80$).

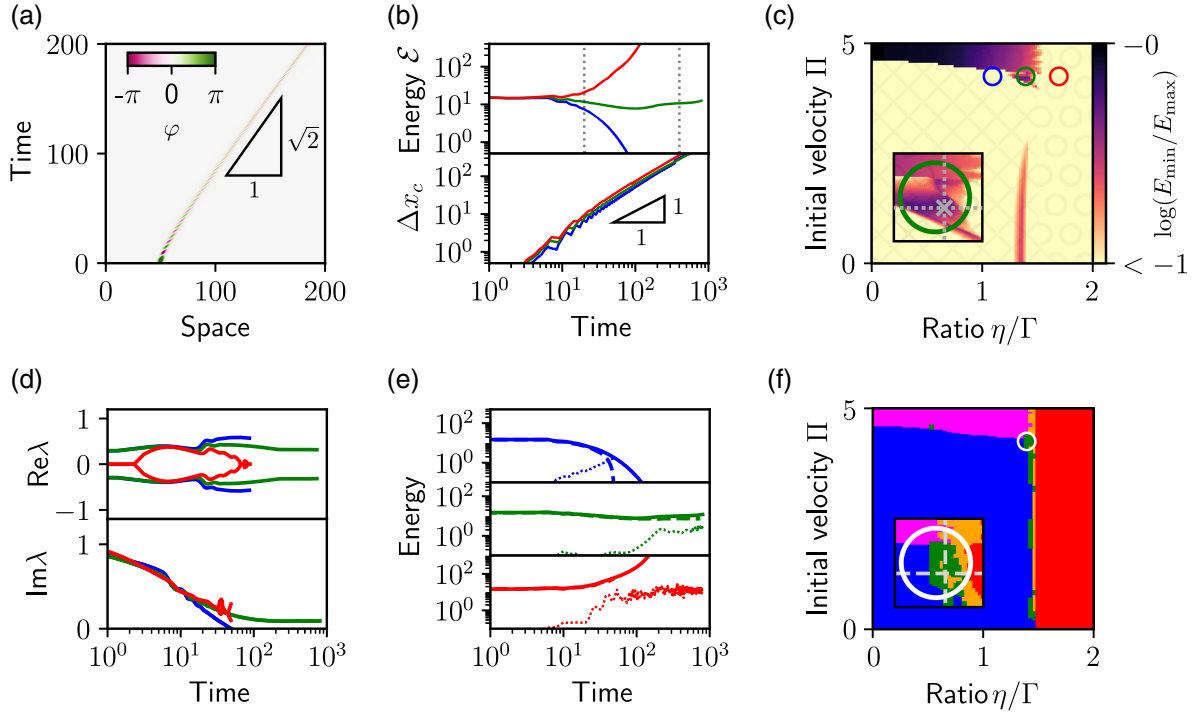


FIG. 3. Breathers in the nonreciprocal sine-Gordon equation. (a) Kymograph of the field φ for $\eta = 0.1373$, $\Gamma = 0.1$, and $\Pi = 4.17$ through a numerical integration of Eq. (2). (b) Total energy $\mathcal{E}$ of the breather (top) as defined by Eq. (4) and the position of breather Δx_c versus time (bottom) for three different values of nonreciprocity, $\eta = 0.10$ (blue line), 0.1373 (green line), and 0.15 (red line). (c) Logarithm of the ratio between the minimum and maximum energy measured in the time interval indicated by the vertical dashed lines in (b) against the nonreciprocity to dissipation ratio η/Γ and the magnitude of the initial kick Π . The energy is defined in Eq. (4). The three colored circles correspond to the three different values plotted in (b). Regions hatched with squares (circles) demarcate the weakly (strongly) nonlinear regime of Eq. (2). Inset: close-up on the region that hosts long-lived breathers. (d) Time dependence of the real (top) and imaginary (bottom) parts of λ_1 and λ_2 for the same values of Π and η . (e) Energy dependence of the soliton profile for the same values of Π and η . Solid lines represent the energy of the field found by numeric integration Eq. (4), and dashed (dotted) lines are contributions from the discrete (radiative) modes only, as given by Eq. (10). (f) Spectral content of the nonlinear wave vs ratio between nonreciprocity and dissipation η/Γ and magnitude of the initial kick Π at time $t = 800$. The narrow green area shows the region of the parameters which correspond to a breather and radiation modes. This area separates decaying profiles consisting solely of radiation modes (blue region) and the regions marked in red and orange that correspond to the diverging profiles and one or more discrete degrees of freedom, respectively. The orange region corresponds to a situation when additionally to a breather other discrete degrees of freedom are present in the profile (i.e., more breathers or, even, kink-antikink pairs). The full energy in this case still should be low enough to compute $a(\lambda)$ and extract the corresponding zeros (contrary to the red region). Note that the “upper” boundary separates breathers from the kink-antikink pairs (orange area). Persistent breathers live only in a tiny region (see inset) of the parameter space $\eta/\Gamma \simeq 1.4$ and $\Pi \simeq 4.2$ indicated by the white circle.

The three regimes of decaying, long-lived and diverging breathers as denoted by the blue, green and red curves in Fig. 3(b), can be distinguished from the scattering data as follows: The dissipative regime is marked by a vanishing imaginary part, while the real part stays finite (blue curve); the exploding regime is marked by the vanishing of both real and imaginary parts (red curve), while in the stable regimes both spectral data evolve slowly over the simulation window (green curve). This confirms the existence of a breathing soliton in the minute region $\eta/\gamma \simeq 1.4$ and $\Pi \simeq 4.2$ [see inset of Fig. 3(c)].

To quantify this behavior even further, we have used the scattering data $a(\lambda)$ to numerically quantify the fraction of energy of discrete structures such as solitons and breathers

in each profile (see Appendix C 3),

$$\mathcal{E}_d = 2 \sum \text{Im}(\lambda_i - \lambda_i^{-1}), \quad (9)$$

which in the case of a single breather becomes

$$\mathcal{E}_b = 2 \text{Im}(\lambda_1 + \lambda_2 - 1/\lambda_1 - 1/\lambda_2). \quad (10)$$

Note that generically more than one breather may appear, together with kinks and antikinks. We can use Eq. (9) to quantify the total energy of such discrete nonlinear structures. In addition, we can quantify the energy of radiation as (see Appendix C 3)

$$\mathcal{E}_r = \int_0^\infty \log \frac{1}{|a(\lambda)|^2} \frac{1}{\pi} d\left(\lambda - \frac{1}{\lambda}\right). \quad (11)$$

Plotting $\mathcal{E}$, $\mathcal{E}_d$, and $\mathcal{E}_r$ vs time for a few different values of parameters [Fig. 3(e)] sheds new light on the three possible scenarios: (i) In the decaying regime (blue curves), the discrete degrees of freedom initially dominate $\mathcal{E}_t \approx \mathcal{E}_d \gg \mathcal{E}_r$, but vanish and are replaced by radiation for long times $\mathcal{E}_t \approx \mathcal{E}_r \gg \mathcal{E}_d$; (ii) in the exploding regime (red curves), we see that the discrete degrees of freedom take the most part of the profile's energy $\mathcal{E}_t \approx \mathcal{E}_d \gg \mathcal{E}_r$; (iii) finally, in the stable regime (green curves), we see that the discrete degrees of freedom dominate the dynamics over a long transient $\mathcal{E}_t \approx \mathcal{E}_d \gg \mathcal{E}_r$. In other regimes for smaller magnitude of initial conditions, the radiation itself spawns discrete structures (see Appendix C 3 for a complete discussion).

By making a snapshot of the spectral content for large times $t = 800$, we obtain a good picture of the nature of the nonlinear waves and identify the regime where breathing solitons persist [Fig. 3(f)]. To obtain the color plot it was enough to deduce the number of zeros of $a(\lambda)$ without having to determine their precise positions. The number of zeros N_C inside a domain bounded by a contour C is given by the following contour integral,

$$N_C = \frac{1}{2\pi i} \oint_C d\lambda \frac{\partial_\lambda a(\lambda)}{a(\lambda)}, \quad (12)$$

providing an efficient way to compute N_C numerically. To compute the number of breathers we have chosen the contour C in the complex plane λ to be a square with the opposite vertices given by $\lambda = 0.1 + 0.1i$ and $\lambda = 1 + 0.7i$. Those values were chosen empirically to encircle most of the zeros. There is a natural limitation for the spectral analysis, as for very large or very small values of λ , the computation of $a(\lambda)$ suffers from numerical instabilities. This can be seen in particular in the energy of the corresponding profiles Eq. (9). Separately we have looked for the zeros on the imaginary line. In terms of the zeros, the colors mean the following: In the blue region there are no zeros; in green there is one with the nonzero real part; for the magenta all zeros have zero real part; in the orange region there are more than two zeros; and the red regions have the energy too high to perform the analysis. For the green and blue areas, in order to ensure that we have found all the zeros, we compared the total energies computed from the profile and with Eqs. (9) and (11).

For low nonreciprocity to dissipation ratio $\eta/\Gamma < 1.4$, we see that at long times, radiation dominates (blue region), except for large value of the kick magnitude $\Pi > 4.4$, where a pair of kink and antikink dominates (magenta region) [Fig. 3(f)]. This regime cannot be achieved in our experiments since the oscillators are monostable. For large nonreciprocity to dissipation ratio $\eta/\Gamma > 1.5$, at long times,

multiple breathers and/or kinks dominate (red region). Only for intermediate nonreciprocity $\eta/\Gamma \simeq 1.4$ will a single breather dominate the dynamics [green region and inset of Fig. 3(f)].

We have seen that breathing solitons emerge for a broad range of initial conditions but persist only for a thin range of nonreciprocity to dissipation ratio η/Γ . What controls this range of persistence and why is it so thin? To address these questions we will next employ a multiple-scale expansion to obtain analytical predictions for the dynamics of the envelope of such nonreciprocal breathing solitons.

IV. ENVELOPE OF THE NONRECIPROCAL BREATHING SOLITON

The adiabatic evolution of the spectral data for the perturbed sine-Gordon equation (2) can, in principle, be analyzed via the inverse scattering perturbation theory [28,72,75,76]. However, this analysis involves a system of four integro-differential equations and at first glance is hardly simpler than the original equation. Therefore, prompted by our observations of Figs. 1 and 3 we extract qualitative predictions by turning to a simpler model of a small-amplitude breather of the form

$$\phi(x, t) = 2\tilde{\psi}(x, t)e^{2it} + \text{c.c.}, \quad (13)$$

where $\psi(x, t)$ is a slowly varying envelope $|\partial_t \psi| \ll |\psi|$, and for convenience we have rescaled time $t \rightarrow t/2$. After this rescaling Eq. (2) takes a canonical form [77] of nonreciprocal dissipative nonlinear Schrödinger equation (NLS):

$$i\partial_t \psi + \psi_{xx} + 2|\psi|^2 \psi = \eta \psi_x - i\Gamma \psi + \frac{\Gamma}{2} \partial_t \psi. \quad (14)$$

This equation assumes that cubic nonlinearities are sufficient to capture the nonlinear dynamics. Our calibrations done in Appendix A [Fig. 6(b)] prove that indeed cubic nonlinearities qualitatively capture range of deformations accessible experimentally. In the absence of perturbations on the right-hand side, Eq. (14) is integrable and the soliton ψ_s describes the envelope of the breather in the laboratory frame,

$$\psi_s(x, t) = \frac{ue^{iv[x-x_c(t)]+i\Phi(t)}}{\cosh\{u[x-x_c(t)]\}}, \quad (15)$$

with phase $\Phi(t)$ and center mass coordinate $x_c(t)$. In integrable dynamics these variables are linear functions of time and are named angle variables, while $u(t)$ and $v(t)$ are constants (the action variables) representing the amplitude and velocity of the envelope. In contrast, in the nonintegrable case nonreciprocity and dissipation will result in more complicated time dependence for these parameters.

We now predict how the dynamics of this soliton—the envelope of the breather—depends on nonreciprocity, dissipation, and initial conditions. To this end, we note that the scattering data are now given by a zero of $a(\lambda)$ in the upper half-plane $\lambda_0 = v + iu$, where the parameters v and u are the velocity and amplitude of the profile Eq. (15). The construction of the transfer matrix is similar to the sine-Gordon case; however, here we do not have any restrictions on the zeros (cf. Appendixes C 2 and C 4). The time evolution of these parameters as well as the parameters Φ and x_c can be obtained with the inverse scattering perturbation theory, which in the adiabatic approximation leads to (see Appendix C 4)

$$\frac{dv}{dt} = \frac{2u^2}{3}(\eta - \Gamma v), \quad (16)$$

$$\frac{du}{dt} = 2u(\eta v - \Gamma) + u\Gamma(u^2 - v^2), \quad (17)$$

and for the angle variables,

$$\frac{dx_c}{dt} = 2v, \quad \frac{d\Phi}{dt} = u^2 + v^2. \quad (18)$$

Although they are impossible to solve analytically, Eqs. (16) and (17) reveal a physically appealing picture. First, there are two fixed points: one at $u = 0$, which corresponds to no solitons at all, and the other at $u = \infty$ and $v = \eta/\Gamma$, which correspond to a diverging soliton. Second, in the regime $|\eta|/\Gamma \leq \sqrt{2}$ there is another pair of fixed points P that corresponds to a soliton of constant velocity with a constant amplitude:

$$v_P = \frac{\eta}{\Gamma}, \quad u_P = \pm \sqrt{2 - \frac{\eta^2}{\Gamma^2}}. \quad (19)$$

To determine which fixed point governs the dynamics, we show the phase portrait in Figs. 4(a)–4(c) and observe that the fixed point at $u = 0$ is stable and attracts all the trajectories if the initial condition sits below the separatrix S defined by

$$u_S^2 = 2 - \eta^2/\Gamma^2 + \frac{3}{5}(v + \eta/\Gamma)^2 \quad (20)$$

and shown as orange curves. In the case $|\eta|/\Gamma \leq \sqrt{2}$, if the initial conditions sit above the separatrix S , the trajectory starts to be attracted toward the fixed point P [Figs. 4(a) and 4(b)]. The envelope of the breather shrinks in magnitude toward a magnitude u_P and its velocity converges toward v_P . Yet this point is unstable and, unless we have started directly from the separatrix S , the magnitude of the envelope ultimately diverges. In the case $|\eta|/\Gamma > \sqrt{2}$, the fixed point P has disappeared and the dynamics again either blows up if the initial condition sits above the

separatrix or dies down if the initial condition sits below [Fig. 4(c)].

There is hence a saddle-node bifurcation at $|\eta|/\Gamma = \sqrt{2}$ at which the fixed point P disappears [Fig. 4(d)]. The soliton never persists because its dynamics is controlled by an unstable fixed point except precisely at this bifurcation, where the timescale—hence the lifetime of the soliton—diverges.

To confirm these findings, we solve Eqs. (16) and (17) numerically and find the two regimes reported above depending on the ratio between nonreciprocity and dissipation η/Γ [Figs. 4(e) and 4(f)] and on the magnitude of the initial condition [Figs. 4(g) and 4(h)]: (i) Above the separatrix $\eta/\Gamma = 1.50$, the soliton accelerates, decays in a short transient before blowing up (red curve); (ii) below the separatrix $\eta/\Gamma = 1.00$, the soliton accelerates until a terminal velocity while its magnitude decreases (blue curve); (iii) finally, close to the separatrix $\eta/\Gamma = 1.373$, the soliton accelerates, its magnitude decreases, but after a transient blows up (green curve). The closer to the separatrix and the closer to the bifurcation, the longer the transient. We further confirm these results with numerical solutions to Eq. (14). The results are consistent even though additional radiations in Eq. (14) induce a quantitative discrepancy with the solutions of Eqs. (16) and (17). Such discrepancy is natural given that these equations describe a single breathing soliton, in the absence of additional waves such as radiations (see Appendix C 4 for a discussion on the role of such radiations).

How do these predictions compare to the numerical solutions of Eq. (2)? We readily see that the ratio η/Γ for which the longest living breather is $\eta/\Gamma \simeq 1.4$ is consistent with the prediction $\eta/\Gamma = \sqrt{2}$ above. The theory also predicts a breather velocity of $\sqrt{2}$, which is fully consistent with the simulation shown in Fig. 3(a) [78]. What about the initial condition? Given that the initial conditions used to numerically solve Eq. (2) are that of a static breather [Eq. (8)], the correspondence between the parameters Π , d , $u(0)$, and $v(0)$ is the following: $u(0) = (\Pi d - 2)/(2d)$ and $v(0) = 0$. Therefore, the optimal initial condition to sit on the separatrix is $u(0) = u_S = \sqrt{2}\sqrt{1 - (\eta/\Gamma)^2/5}$, which corresponds to $\Pi_S = [2 + 2d\sqrt{2}\sqrt{1 - (\eta/\Gamma)^2/5}]/d = 2\sqrt{6/5} + 2 \simeq 4.19$ in the limit case $\eta/\Gamma = \sqrt{2}$ —we chose $d = 1$ for our simulations. This prediction is in beautiful agreement with our simulations, where we found that the optimal initial conditions to observe the longest breathers were $\Pi = 4.2$.

In conclusion, we now understand why the region of existence of the breather is so narrow: Its dynamics is underpinned by an unstable fixed point, which leads to a long-lived soliton if two conditions are met. (i) While this fixed point exists for a wide range of parameters, only when the ratio between nonreciprocity and dissipation is $\eta/\Gamma = \pm\sqrt{2}$ will this pair of fixed points undergo a bifurcation

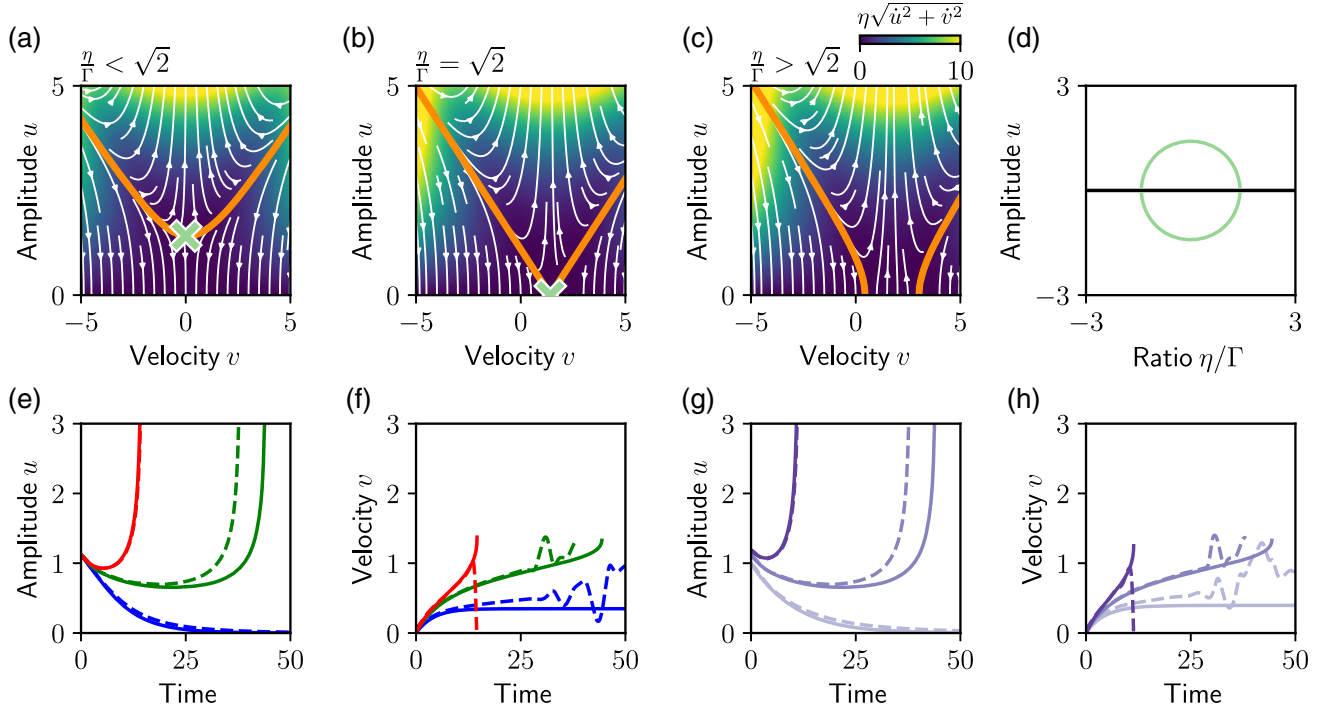


FIG. 4. Longtime dynamics of the envelope of the soliton. (a) Phase portrait u vs v for $\eta/\Gamma = 0$. White streamlines and color map indicate the direction and rate of flow, respectively. The orange curve shows the separatrix given by Eq. (20) and the light green cross indicates the fixed point given by Eq. (19). (b) Phase portrait u vs v for $\eta/\Gamma = \sqrt{2}$ and (c) phase portrait u vs v for $\eta/\Gamma = \sqrt{3}$ from Eqs. (16) and (17). (d) Bifurcation diagram u vs η/Γ . The light green line indicates the pair of fixed points P and the black line corresponds to the fixed point at $u = 0$. (e), (f) Trajectories u (e) and v (f) vs time for $\eta = 0.15$ (red line), 0.1373 (green line), and 0.1 (blue line) and $\Gamma = 0.1$. Solid lines indicate numerical solutions of Eqs. (16) and (17) and dashed lines numerical solutions of Eq. (14) with Eq. (15) as an initial condition with $\Phi(0) = 0$, and $u = 1.12$. (g), (h) Trajectories u (g) and v (h) vs time for $u = 1.19$ (dark purple line), 1.12 (purple line), and 1.0 (light purple line) and $\Gamma = 0.1$ and $\eta = 0.1373$.

and the timescale of the transient diverge; (ii) only when the initial conditions are precisely tuned close to the separatrix will the dynamics approach this fixed point.

These results are seemingly at odds with the experimental and numerical observations reported in Fig. 2, where long-lasting breathing solitons were also observed for a wide range of initial conditions. To resolve this issue, we run simulations of Eq. (1) with varying degrees of discreteness D (see Appendix B 1 and Fig. 7). We observe that as the discreteness parameter D decreases, viz., as we tend toward the continuum limit, the region of stability progressively shrinks and converges toward a point. Hence, in the discrete case, the line of stability appears to be a vestige of the island of stability we predicted analytically previously in this section, but discreteness enhances stability.

V. EFFECT OF DISCRETENESS

There is a powerful way to address the role of discreteness in the NLS model. Indeed, contrary to the Frenkel-Kontorova model discussed in Sec. II, the NLS equation (14) admits an integrable discretization known as the Ablowitz-Ladik (AL) model [79–81]. Following the ideas presented in Ref. [82], we therefore focus on the discrete

form of Eq. (14), given by

$$i\partial_t \psi_n + \frac{\psi_{n+1} + \psi_{n-1} - 2\psi_n}{a^2} + |\psi_n|^2(\psi_{n+1} + \psi_{n-1}) = R_n[\psi] + R_n^{(d)}[\psi]. \quad (21)$$

Here, the left-hand side represents the aforementioned AL discretization, with a denoting the lattice spacing. This system possesses an exact one-soliton solution, parametrized by constants u and v , of the form

$$\psi_n^{(s)}(t) = \frac{\sinh(au)e^{ivna+i\Omega(t)}}{a \cosh\{u[na - x_c(t)]\}}, \quad (22)$$

where the soliton center and phase evolve as

$$\frac{dx_c}{dt} = \frac{2 \sin(av) \sinh(au)}{a^2 u}, \quad (23)$$

$$\frac{d\Omega}{dt} = \frac{2[\cos(av) \cosh(au) - 1]}{a^2}. \quad (24)$$

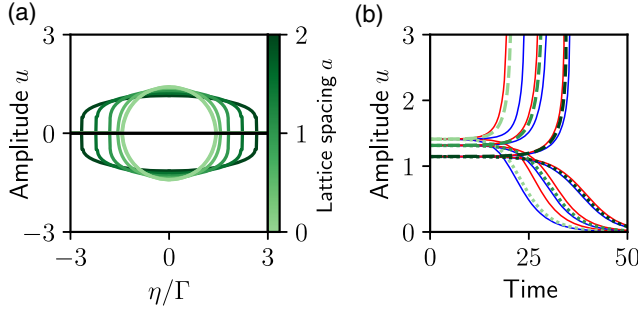


FIG. 5. Discrete effects in the nonlinear Schrödinger equation. (a) Bifurcation diagram as a function of the lattice spacing a . The light green line corresponds to the fixed points depicted in Fig. 4(d). (b) Time evolution of the amplitude according to Eqs. (27) and (28), with initial conditions (u, v) corresponding to the system's fixed point but with the u coordinate perturbed in the positive (negative) direction by 0.3%, shown in the dashed (dotted) lines, for three lattice spacings $a = [0.1, 1, 2]$ and with $\eta = 0.01$ and $\Gamma = 0.1$. Red and blue lines surrounding the green lines indicate the effect of an additional perturbation in η by $\pm 0.015\%$.

In the continuum limit $a \rightarrow 0$, this discrete solution converges to the continuous profile Eq. (15). The perturbation terms $R_n(\psi)$ are defined by the physical damping and nonreciprocity,

$$R_n[\psi] = \eta \frac{\psi_{n+1} - \psi_n}{a} - i\Gamma\psi_n + \frac{\Gamma}{2}\partial_t\psi_n, \quad (25)$$

while the terms $R^{(d)}[\psi]$ measure the difference between the naive discretization of the NLS model and the AL discretization [82]:

$$R_n^{(d)}[\psi] = |\psi_n|^2(\psi_{n+1} + \psi_{n-1} - 2\psi_n). \quad (26)$$

The evolution of the soliton parameters u and v can be derived using inverse scattering perturbation theory, as in the continuous case (see also Ref. [83]), or alternatively through the analysis of conserved quantities. This yields

$$\frac{du}{dt} = X_a^{(u)}(u, v), \quad (27)$$

$$\frac{dv}{dt} = X_a^{(v)}(u, v) - \frac{(2\pi)^3 \sinh(au)^2}{a^6 u^3} \sum_{n=1}^{\infty} \frac{n^2 \sin(2\pi n x_c/a)}{\sinh \frac{\pi^2 n}{ua}}, \quad (28)$$

with the explicit expressions for the vector fields $X_a^{(u,v)}$ provided in Appendix C [Eqs. (C44) and (C45)].

In the limit $a \rightarrow 0$, these expressions reduce to the continuum dynamics Eqs. (16) and (17). The final term in Eq. (28) represents the Peierls-Nabarro (PN) contribution, arising from the lattice structure. In our analysis, we

retain only the $n = 1$ term in this sum, an approximation also performed in Ref. [82], and neglect the corresponding PN contributions to $X_a^{(u,v)}$, as they are small not only due to discreteness, but also due to the perturbations. This way, $X_a^{(u,v)}$ define again a two-dimensional flow, qualitatively similar to that shown in Figs. 4(a)–4(c). However, the bifurcation point now depends on the lattice spacing through the modified condition $|\eta|/\Gamma \leq \sqrt{2 + a^2}$, which flattens the bifurcation line of Fig. 4(d) [see Fig. 5(a)]. Hence, the sensitivity to the magnitude of the initial condition is weaker, in accordance with our findings of Fig. 7. To better isolate the role of lattice spacing, we focus on the regime of small η , where the dependence of the bifurcation point on a is minimal. It is important to note that the inclusion of the PN term distorts the simple 2D flow structure. We therefore compute the critical points numerically and analyze the perturbations around them. The corresponding results shown in Fig. 5(b) demonstrate that discreteness enhances soliton lifetime but also renders the dynamics less sensitive to changes in the nonreciprocal parameter η .

VI. DISCUSSION

The existence of these genuine breathing solitons in nonreciprocal nonlinear media is in stark contrast with that of solitons in constantly driven media, which admit static short-lived breathers [43–45], yet whose behavior does not seem to correspond to any fixed point for reasonable values of the driving (see Appendix C 5). We emphasize here that our considerations for the soliton solution in the nonreciprocal nonlinear Schrödinger equation (14) serve as an enveloping profile for the breather of the nonreciprocal sine-Gordon equation (2). In contrast, the breathers in Refs. [43–45] from the point of view of the driven nonlinear Schrödinger equation correspond to the special solutions of Kuznetsov-Ma type [42,84] that could be obtained by a dressing or degeneration of the periodic solutions [84]. In addition, other types of static breathing solitons, sometimes called oscillons [85], have a more complicated solution to the nonlinear sine-Gordon equation called Dashen, Hasslacher, and Neveu (DHN) solution for which an inverse scattering analysis is impossible. Crucially, both these types of standing breathers are static, in contrast to our nonreciprocal breathers that travel.

We interpret the difference between these types of breathers as follows. A breathing soliton can be seen as a bound pair of kink and antikink that oscillates in time. In ordinary driven media, kinks and antikinks are driven toward opposite directions [40]. This means that the driving tends to keep the breather from traveling and to pull the kink and antikink apart, thereby to disintegrate the breather. In contrast, in nonreciprocal media, kinks and antikinks are driven toward the same direction [28]. This means that the nonreciprocal driving has the tendency to drive the breather

unidirectionally and to maintain its shape. Our results here have demonstrated that such kink-antikink pair making up the breather is, however, a delicate balance and persists only if the energy injection precisely compensates the dissipation and if the initial conditions have triggered a breather with the suitable magnitude.

In conclusion, we have discovered how nonreciprocity affects the emergence and longtime dynamics of breathing solitons in a nonreciprocal active material. Our exploration has been performed experimentally, numerically, and theoretically for nonreciprocal generalizations of the sine-Gordon and nonlinear Schrödinger equations. In the generic case, nonreciprocity amplifies localized initial conditions toward unidirectional nonlinear structures such as radiations, single or multiple breathers, or kink-antikink pairs. Such nonlinear excitations generically either are damped or diverge over short timescales. Only for a specific region of parameter space, when nonreciprocal gain, dissipation, and initial magnitude are carefully balanced, will a pure breathing soliton emerge, propagate unidirectionally, and remain stable over a long timescale. This careful balance corresponds to the bifurcation of a pair of unstable fixed points, which can be reached by the dynamics provided the initial conditions are precisely tuned. In addition to these continuum theoretical results, we have extended our analysis to a discrete setting, where we found that discreteness has a stabilizing effect and tends to expand the regime of existence and lifetime of nonreciprocal breathers, which explains our experimental and numerical observations.

We have identified a minimal set of ingredients necessary for nonreciprocal breathers to emerge: Weakly nonlinear effects must stabilize otherwise dispersive waves while nonreciprocal forces counteract dissipation to sustain breather solitons. Nonreciprocity itself may arise in a variety of ways. In the current work, these interactions are explicitly programmed and applied using electrical power source, but they arise generically in systems with asymmetric phase delays [14,86], asymmetric exchange interactions [87], mediated by hydrodynamic fields [18–20,22,88], acoustic streaming [89,90], or light fields [17,91]. In the majority of these systems, mechanical oscillations and waves can exist and, hence, these platforms are candidates to potentially host nonreciprocal breathers.

Our work establishes nonreciprocity as a mechanism of choice to obtain traveling breathing solitons in active or dissipative materials and, hence, to transport information and energy in dissipative media. Open questions ahead are how to increase further the range of stability of such breathers and to explore their stability to disorder and noise.

ACKNOWLEDGMENTS

We thank Daan Giesen and Tjeerd Weijers for technical support. We thank Filip Novkoski and Stephane Randoux for discussion on the numerical implementation

of the inverse scattering method. We acknowledge funding from the European Research Council under Grant Agreement No. 852587 and from the Netherlands Organisation for Scientific Research under Grant Agreement No. VI.Vidi.213.131. M.B. acknowledges funding from the European Research Council under Grant Agreement No. 101117080.

DATA AVAILABILITY

The data that support the findings of this article are openly available [92].

APPENDIX A: EXPERIMENTAL METHODS

Figure 1(a) shows four coupled oscillators that are part of an active metamaterial consisting of 50 identical unit cells, separated by a lattice spacing of $a = 6$ cm. The unit cells are directly inspired by Ref. [28]. Each oscillator arm has a rotational degree of freedom θ_i with moment of inertia $I = 3 \times 10^{-5}$ kg m² and is coupled elastically to its neighbors by 2 rubber bands of length 4.6 cm and thickness 2 mm. As the difference in angle between two neighbors $\theta_i - \theta_{i+1}$ changes, the rubber bands induce a torque $\tau_i = \kappa(\theta_i - \theta_{i+1})$ where $\kappa = 7.5$ mN m/rad is the torsional stiffness that we calibrate using a torsion testing machine [Instron E3000 linear-torsional; see Fig. 6(a)]. Besides these passive elastic interactions that are symmetric with respect to left and right neighboring sites $i - 1$ and $i + 1$, each unit incorporates an electromechanical motor implementing an active feedback that breaks this symmetry. Each unit applies a torque

$$\tau_i^a = \kappa^a(\theta_{i-1} - \theta_{i+1}), \quad (\text{A1})$$

where κ^a represents the nonreciprocal interaction strength. Angular positions are measured and recorded by an angular encoder (Broad-com HEDR-55L2-BY09) present in each unit cell. This information is transmitted between nearest neighbors at a rate of 100 Hz and used to compute the torque τ_i^a to be exerted by a direct-current coreless motor (Motraxx CL1628) via a microcontroller (Espressif ESP 32), both of which are embedded in the unit cell. Each oscillator arm holds a neodymium magnet which interacts with an identical magnet placed regularly on a substrate. Contrary to the setup of Ref. [28], where substrate magnets are spaced alternately in between oscillators, here we align magnets directly opposite each oscillator arm. This generates a weakly nonlinear monostable magnetic potential that induces a torque that we measure to be approximately sinusoidal with the oscillator angle θ_i and has an amplitude of $B = 2.3$ mN m [see Fig. 6(b)]. The viscous dissipation associated to oscillator rotation was found to be $\gamma = 0.17$ mN m s/rad by fitting the oscillation amplitude decay after an initial perturbation [see Fig. 6(c)].

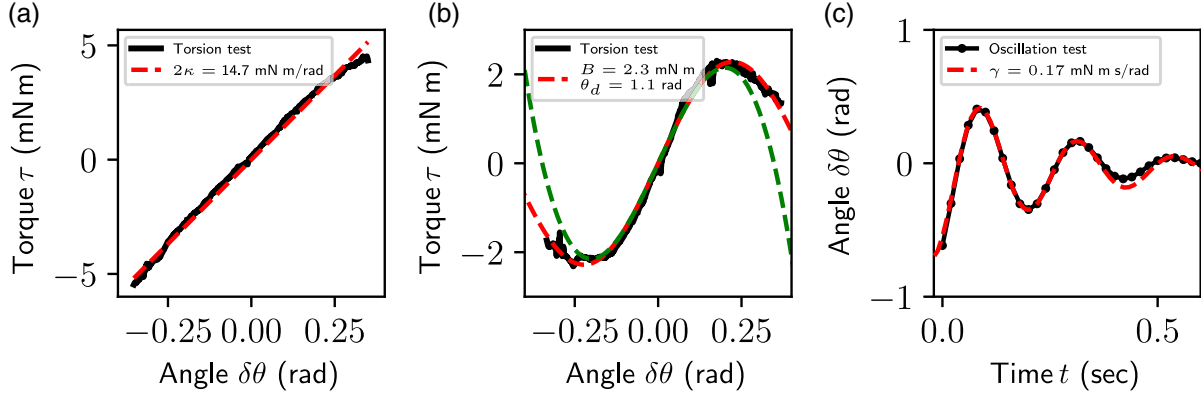


FIG. 6. Calibration of experimental model parameters. (a) Torque-angle curve measuring the elastic response of a single unit without the magnetic and nonreciprocal contributions. While keeping the angles of the two neighboring oscillators fixed, the elastic torque generated is measured with an Instron E3000 linear-torsional testing machine. A linear fit yields $\kappa \approx 7.5$ mNm/rad. (b) Torque-angle curve measuring the magnetic response of a single unit without the elastic and nonreciprocal contributions. A sinusoidal fit yields amplitude $B \approx 2.3$ mNm and period $\theta_d \approx 1$ rad. The dashed green line indicates the third order Taylor expansion of the fitted sine function, which validates the weakly nonlinear approximation done in the main text. (c) Decaying oscillation after perturbation of a single oscillator. We fit the data to a decaying sine function $\propto \exp(-\gamma t) \cos(\omega t)$ to recover the damping coefficient $\gamma = 0.17$ mNm s/rad.

For an appropriate nonreciprocal interaction strength κ^a , the perturbation sets in motion the breathing pulse shown in Fig. 1(c). Using the total squared amplitude as a measure for the system's energy $E = \sum_i \theta_i^2$, we quantify the breather's lifetime by measuring the ratio between minimum and maximum energy within a time window of one starting shortly after the initial kick has ringed down. This ratio will be close to 1 for waves whose energy does not change significantly over time, while it vanishes for decaying and amplifying waves. In Fig. 2(b), we plot the logarithm of this quantity for a sweep over initial velocities and nonreciprocal coupling strengths. Note that alternative choices for the time window change only the scale of the color map but preserve the observed bifurcation between decaying and amplifying waves.

APPENDIX B: NUMERICAL METHODS

1. Nonreciprocal Frenkel-Kontorova model

We model the dominant forces—inertial, elastic, viscous, nonreciprocal, and magnetic—present in the active metamaterial with a nonreciprocal version of the Frenkel-Kontorova model:

$$I \frac{\partial^2 \theta_i}{\partial \tau^2} = \kappa(\theta_{i-1} + \theta_{i+1} - 2\theta_i) - \kappa^a(\theta_{i+1} - \theta_{i-1}) - \gamma \frac{\partial \theta_i}{\partial \tau} + B \sin\left(2\pi \frac{\theta_i}{\theta_d}\right), \quad (\text{B1})$$

where τ denotes dimensionful time and $\theta_d = 1$ rad is the width of the magnetic potential, found by fitting the measured magnetic field to a sine function [see Fig. 6(b)]. By employing the following substitutions, we find the

nondimensional form of Eq. (1):

$$\begin{aligned} \phi_i &= 2\pi \frac{\theta_i}{\theta_d}, \\ t &= \sqrt{\frac{\kappa}{I}} \tau, \\ \eta &= \frac{2\kappa^a}{\kappa\sqrt{D}}, \\ \Gamma &= \frac{\gamma}{\sqrt{IkD}}, \\ D &= \frac{2\pi B}{\kappa\theta_d}. \end{aligned} \quad (\text{B2})$$

We note that this model is identical to the one discussed in Ref. [28] in the context of topological solitons, but we focus here instead on the weakly nonlinear regime where oscillators do not hop between different minima. To compare model to experiment, we integrate Eq. (1) with the velocity Verlet algorithm for a system of $N = 50$ oscillators using the experimentally calibrated parameters and time step $dt = 0.1$. Mirroring the experiment, in Figs. 2(c) and 2(d), a single site is initialized with a nonzero angular velocity $\dot{\theta}_{t=0}$ and stress-free boundary conditions are imposed.

In Fig. 7, we plot the breather stability for $D = \{3.6, 1.0, 0.3\}$ and observe that as D decreases and the continuum limit is approached, the stable region depends not only on the activity η , but also on the amplitude of the initial condition Π . This confirms our finding that breathers in the continuum descriptions discussed in Figs. 3 and 4 of the main text, are stable only when both activity and initial condition are tuned. These findings also suggest that discreteness enhances breather stability in nonreciprocal systems.

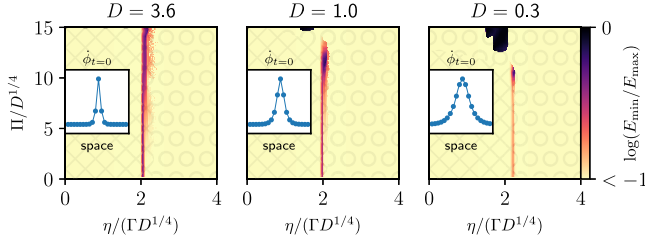


FIG. 7. Stability of breathers with decreasing discreteness. Phase diagram of the logarithm of the ratio between minimum and maximum energy $\sum \theta_i^2$ visited by the system up until nondimensional time $t = 1000$. As D decreases, the vertical stable region also seen in Figs. 2(b) and 2(d) morphs into a pointlike region as seen and predicted in Fig. 3(c). The black region in the $D = 0.3$ case corresponds to topological solitons, also observed in Fig. 3(c). Regions hatched with squares (circles) demarcate the weakly (strongly) nonlinear regime of Eq. (2). Insets show the shape of the initial condition $\phi_{t=0}$.

2. Simulation of continuum equations

Simulations of both the nonreciprocal sine-Gordon and nonlinear Schrödinger equations, shown in Figs. 3 and 4, respectively, were performed using the open source package py-pde [93] with adaptive time step and 601 spatial discretization steps. For Figs. 3(b) and 3(c), the initial condition corresponds to the breather profile described by Eq. (3), while Fig. 4 uses Eq. (15) as an initial condition with $\Phi(0) = 0$.

APPENDIX C: THEORY

1. Continuum limit of the discrete model

The continuum limit of Eq. (1) is found by letting ϕ_i become a continuous function $\phi(x)$ of space $x \in [0, Na]$, with N the number of units. We approximate finite differences by a Taylor expansion according to $\phi_{i+1} - \phi_i \approx a\phi_x + a^2\phi_{xx}/2$ and substituting terms in the discrete model of Eq. (1) then leads to Eq. (2) under rescaling of the spatial variable $x \rightarrow (a/\sqrt{D})x$ and time $t \rightarrow (t/\sqrt{D})$.

2. Inverse scattering method for the sine-Gordon equation

In this section, we briefly review the main points of the inverse scattering method for solving classical integrable (1+1)-dimensional field theories. We follow mainly Ref. [73]. We start with the sine-Gordon equation:

$$\partial_t^2 \varphi - \partial_x^2 \varphi + \sin \varphi = 0. \quad (C1)$$

The key observation is that this equation is equivalent to the following *auxiliary* linear system,

$$\frac{\partial F}{\partial x} = UF, \quad \frac{\partial F}{\partial t} = VF, \quad (C2)$$

where U and V are 2×2 matrices that depend on the field φ and the auxiliary spectral parameter λ :

$$U = \frac{\partial_t \varphi \sigma_3}{4i} + \frac{\lambda + \lambda^{-1}}{4i} \sigma_1 \sin \frac{\varphi}{2} + \frac{\lambda - \lambda^{-1}}{4i} \sigma_2 \cos \frac{\varphi}{2}, \quad (C3)$$

$$V = \frac{\partial_x \varphi \sigma_3}{4i} + \frac{\lambda - \lambda^{-1}}{4i} \sigma_1 \sin \frac{\varphi}{2} + \frac{\lambda + \lambda^{-1}}{4i} \sigma_2 \cos \frac{\varphi}{2}. \quad (C4)$$

The equivalence to Eq. (C1) comes from the compatibility condition of these linear systems (also called zero-curvature condition),

$$\frac{\partial}{\partial x} \frac{\partial}{\partial t} F = \frac{\partial}{\partial t} \frac{\partial}{\partial x} F, \quad (C5)$$

or equivalently,

$$\partial_t U - \partial_x V + [U, V] = 0. \quad (C6)$$

Demanding that this relation is satisfied for every value of λ we obtain our original equation (C1).

The important part of the scattering formalism are Jost solutions $T_{\pm}(x, \lambda)$. They are defined as solutions of the linear system Eq. (C2) at the fixed moment of time,

$$\frac{dT_{\pm}(x, \lambda)}{dx} = UT_{\pm}(x, \lambda), \quad (C7)$$

and are specified by the following asymptotic behavior:

$$T_{-}(x \rightarrow -\infty) = \mathcal{E} e^{-ix\sigma_3(\lambda^2-1)/(4\lambda)} \equiv E(x, \lambda), \quad (C8)$$

$$T_{+}(x \rightarrow +\infty) = \begin{cases} (-1)^{Q/2} E(x, \lambda) & Q \text{ even} \\ (-1)^{(Q-1)/2} i\sigma_3 E(x, \lambda) & Q \text{ odd}, \end{cases} \quad (C9)$$

where matrix $\mathcal{E}$ and topological charge Q are given by

$$\mathcal{E} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}, \quad Q = \frac{1}{2\pi} \int_{-\infty}^{\infty} dx \partial_x \varphi(x). \quad (C10)$$

The reason behind these definitions is twofold. First, we consider very specific solutions of the sine-Gordon equation, that at $x \rightarrow \pm\infty$ approach constant value $\partial_t \varphi(x \rightarrow \pm\infty) = 0$; moreover, these values can differ by the topological charge:

$$\varphi(+\infty) - \varphi(-\infty) = \frac{1}{2\pi} \int dx \partial_x \varphi = Q. \quad (C11)$$

In particular, we can put $\varphi(-\infty) = 0$ so the function $E(x, \lambda)$ is a solution of the linear problem with the asymptotic matrix $U(x \rightarrow -\infty)$. The second reason is more subtle since obviously the solutions of the linear problem Eq. (C7) are defined up to multiplication on the constant

matrix from the right. The choice of this matrix that leads to the definition Eq. (C8) is dictated by the analytic properties of the Jost solutions as functions of the spectral parameter λ ; namely, one can show that similar to the columns of $E(x, \lambda)$, the columns $T_{\pm}^{(l)}$, $l = 1, 2$ of the Jost matrices $T_{\pm}(x, \lambda)$ have the following analytical properties: $T_{\pm}^{(1)}$ and $T_{+}^{(2)}$ can be analytically continued to the upper half-plane, and $T_{-}^{(2)}$ and $T_{+}^{(1)}$ to the lower half-plane of the complex variable λ . Moreover, the specific form of the differential equation allows for the following conjugational properties (involutions):

$$\bar{T}_{\pm}(x, \lambda) = \sigma_2 T_{\pm}(x, \lambda) \sigma_2, \quad \bar{T}_{\pm}(x, -\lambda) = -i \sigma_1 T_{\pm}(x, \lambda). \quad (\text{C12})$$

The Jost solutions should be related by the constant matrix multiplied from the right. Traditionally, this matrix is called the transfer matrix:

$$T_{-}(x, \lambda) = T_{+}(x, \lambda) T(\lambda), \quad T(\lambda) = \begin{pmatrix} a(\lambda) & -\bar{b}(\lambda) \\ b(\lambda) & \bar{a}(\lambda) \end{pmatrix}. \quad (\text{C13})$$

Since the matrix U is traceless, the determinant of Jost solutions does not change with time $\partial_t \det T(x) = 0$, which means the unimodularity of the transfer matrix $T(\lambda)$:

$$\det T(\lambda) = |a(\lambda)|^2 + |b(\lambda)|^2 = 1. \quad (\text{C14})$$

The form of the transfer matrix $T(\lambda)$ follows from the first of the involutions Eq. (C12), which implies

$$\bar{T}(\lambda) = \sigma_2 T(\lambda) \sigma_2. \quad (\text{C15})$$

The second involution implies

$$\bar{T}(\lambda) = T(-\lambda), \quad (\text{C16})$$

which is equivalent:

$$a(-\lambda) = \bar{a}(\lambda), \quad b(-\lambda) = \bar{b}(\lambda). \quad (\text{C17})$$

Moreover, we can express

$$a(\lambda) = \det(T_{-}^{(1)}, T_{+}^{(2)}); \quad (\text{C18})$$

therefore, $a(\lambda)$ can be analytically continued to the upper half-plane. The bound states in the scattering are described by zeros in the upper half-plane, which due to Eq. (C17) can be either purely imaginary $\lambda = i\kappa_n$, $\kappa_n > 0$ or come in pairs $\lambda = \lambda_n$, $\lambda = -\bar{\lambda}_n$, $\text{Im } \lambda_n > 0$ [see Eq. (C12)]. Moreover, using Kramers-Kronig relations along with Eq. (C14) one can present

$$a(\lambda) = R(\lambda) \prod_{n=1}^{N_1} \frac{\lambda - i\kappa_n}{\lambda + i\kappa_n} \prod_{k=1}^{N_2} \frac{\lambda - \lambda_k}{\lambda - \bar{\lambda}_k} \frac{\lambda + \bar{\lambda}_k}{\lambda + \lambda_k}, \quad (\text{C19})$$

where $R(\lambda)$ is referred to in the main text as the radiative part and is given explicitly as

$$R(\lambda) = \exp \left(\int \frac{\ln(1 - |b(\mu)|^2)}{\mu - \lambda - i0} \frac{d\mu}{2\pi i} \right). \quad (\text{C20})$$

From the presentation of Eq. (C18), we see that whenever the spectral parameter coincides with one of the zeros of $a(\lambda)$, then $\lambda = \lambda_j$, and the first and second columns are proportional to each other:

$$T_{-}^{(1)}(x, \lambda_j) = \mathbf{g}_j T_{+}^{(2)}(x, \lambda_j), \quad a(\lambda_j) = 0. \quad (\text{C21})$$

The proportionality coefficient $\mathbf{g}_j$ is another scattering data along with the coefficient $b(\lambda)$ and the position of zeros λ_j . The time evolution of the scattering data can be deduced from the second equation in Eq. (C2) or the zero-curvature condition Eq. (C6):

$$a(\lambda, t) = a(\lambda, 0), \quad b(\lambda, t) = b(\lambda, 0) e^{it(\lambda^2 + 1)/(2\lambda)}, \quad (\text{C22})$$

$$\mathbf{g}_j(t) = e^{it(\lambda_j^2 + 1)/(2\lambda_j)} \mathbf{g}_j(0). \quad (\text{C23})$$

Note that the nonlinear time dynamics of the original equation reduces to the extremely simple evolution on the scattering data. Therefore, the typical way to solve the Cauchy problem for the sine-Gordon equation is to first find the scattering data, i.e., to solve Eq. (C7) (so-called direct scattering problem), then evolve it according to Eq. (C22), afterward restore the potential by means of the inverse scattering problem, i.e., from the given scattering data. The final step typically involves solving linear integral equations to recover the profile. However, for so-called reflectionless potentials, where $b(\lambda) \equiv 0$, these integral equations reduce to algebraic ones—yielding profiles with profound physical significance. In particular, for

$$a(\lambda) = \frac{\lambda - \lambda_1}{\lambda - \bar{\lambda}_1} \frac{\lambda + \bar{\lambda}_1}{\lambda + \lambda_1}, \quad \lambda_1 = \xi + i\nu. \quad (\text{C24})$$

The corresponding profile is given by a breather [see Eq. (5)]. A solution that corresponds to a single imaginary zero,

$$a(\lambda) = \frac{\lambda - i\kappa}{\lambda + i\kappa}, \quad \kappa > 0, \quad (\text{C25})$$

is called a kink (soliton), and has a profile

$$\varphi(x, t) = 4 \arctan \exp \left(\frac{x - vt - x_0}{\sqrt{1 - v^2}} \right). \quad (\text{C26})$$

Contrary to a breather, this solution has a nonzero topological charge $Q = +1$ [see Eq. (C11)]. The antikink is obtained by changing the overall sign in Eq. (C26) and corresponds to $Q = -1$.

As an example of the direct scattering problem, let us discuss a spectral content that corresponds to the following profile:

$$\varphi(x, 0) = 0, \quad \partial_t \varphi(x, 0) = \frac{\Pi}{\cosh(x/d)}. \quad (\text{C27})$$

This profile is used to imitate the initial kick in the experiment. To compute the scattering data we make a substitution $F = \mathcal{E}f$ in the linear system Eq. (C2), rescale $x = 4y/\Pi$, and denote $p = 2(\lambda - 1/\lambda)/\Pi$, which leads to

$$\frac{df}{dy} = \left(\frac{\sigma_3 p}{2i} - \frac{i\sigma_2}{\cosh \frac{4y}{\Pi d}} \right) f. \quad (\text{C28})$$

Taking into account results from Ref. [94] we obtain

$$a(\lambda) = \frac{\Gamma^2\left(\frac{1}{2} - i\frac{(\lambda-1/\lambda)d}{4}\right)}{\Gamma\left(\frac{1}{2} - i\frac{(\lambda-1/\lambda)d}{4} - \frac{\Pi d}{4}\right) \Gamma\left(\frac{1}{2} - i\frac{(\lambda-1/\lambda)d}{4} + \frac{\Pi d}{4}\right)}, \quad (\text{C29})$$

$$b(\lambda) = \frac{\sin \frac{\pi \Pi d}{4}}{\cosh \frac{\pi(\lambda-1/\lambda)\Pi d}{8}}. \quad (\text{C30})$$

So $b(\lambda) = 0$ for $\Pi d = 4n$, $n \in \mathbb{Z}_+$. For instance, $n = 1$ correspond to

$$a(\lambda) = \frac{\lambda^2 - 2i\lambda/d - 1}{\lambda^2 + 2i\lambda/d - 1}. \quad (\text{C31})$$

So for $d < 1$ we have kink-antikink pair and for $d > 1$ we have breather. The special case $\Pi = 4$, $d = 1$ implies the “degenerate” scattering data:

$$a(\lambda) = \left(\frac{\lambda - i}{\lambda + i} \right)^2. \quad (\text{C32})$$

In (x, t) space this corresponds to the following solution:

$$\varphi(x, t) = 4 \arctan \frac{t}{\cosh(x - x_0)}. \quad (\text{C33})$$

If $\Pi d < 4$ we do not have any discrete degrees of freedom.

3. Conserved quantities

Since integrable evolution does not change $a(\lambda)$ its expansion coefficients around some point $\lambda = \lambda_0$ correspond to conserved quantities. It turns out that if we consider expansions around $\lambda = 0$ and $\lambda = \infty$, i.e.,

$$\log a(\lambda) = i \sum_{n=0}^{\infty} I_{-n} \lambda^n, \quad \log a(\lambda) = i \sum_{n=1}^{\infty} \frac{I_n}{\lambda^n}, \quad (\text{C34})$$

the coefficients I_n are given by the integrals of local densities. In particular, I_0 is related to the total charge Eq. (C11), and $I_{\pm 1} \equiv I_{\pm}$ are given by

$$I_{\pm} = \frac{1}{4} \int dx \left(\frac{(\partial_t \varphi \pm \partial_x \varphi)^2}{2} + 1 - \cos \varphi \right). \quad (\text{C35})$$

Note that the energy is presented as a sum of these expressions:

$$\mathcal{E}_t = I_+ + I_-. \quad (\text{C36})$$

Using presentation Eq. (C19) we obtain

$$I_{\pm} = I_{\pm}^{\text{rad}} + I_{\pm}^{\text{d}}, \quad (\text{C37})$$

where the radiation part is given by

$$I_{-}^{\text{rad}} = \int_0^{\infty} \log \frac{1}{|a(\lambda)|^2} \frac{d\lambda}{\pi}, \quad I_{+}^{\text{rad}} = \int_0^{\infty} \frac{1}{\lambda^2} \log \frac{1}{|a(\lambda)|^2} \frac{d\lambda}{\pi}. \quad (\text{C38})$$

The energy of radiation can be presented as

$$\mathcal{E}_r \equiv I_{-}^{\text{rad}} + I_{+}^{\text{rad}} = \int_0^{\infty} \log \frac{1}{|a(\lambda)|^2} \frac{1}{\pi} d\left(\lambda - \frac{1}{\lambda}\right). \quad (\text{C39})$$

The discrete part can be expressed as

$$I_{-}^{\text{d}} = 2 \sum \text{Im } \lambda_i, \quad I_{+}^{\text{d}} = -2 \sum \text{Im } \lambda_i^{-1}. \quad (\text{C40})$$

Here by λ_i we understand every sort of zero in the upper half-plane. This way, every kink is counted as $\lambda_i = i\kappa_i$, while every breather comes with λ_i and $-\bar{\lambda}_i$. By computing $a(\lambda)$ numerically we can quantify the fraction of radiation in each profile. To do this we plot in Fig. 8 the full energy $\mathcal{E}_t$ and the radiation energy $\mathcal{E}_r$. We clearly see three possible scenarios: (i) in the decaying regime (the left-hand column) where discrete degrees of freedom vanish and the profile is determined completely by radiation $\mathcal{E}_r \approx \mathcal{E}_t$; (ii) in the exploding regime (the right-hand column) we see that the discrete degrees of freedom take the most part of the profile's energy $\mathcal{E}_r \ll \mathcal{E}_t$; (iii) finally in the central column we see examples of how discrete degrees of freedom are created from the radiation, i.e., we have the regime (i) followed by a revival, followed by the regime (ii) in the course of evolution. This way, we see that the decay of the solution is possible only if the full profile turns itself completely into the radiation modes, while the amplification is possible

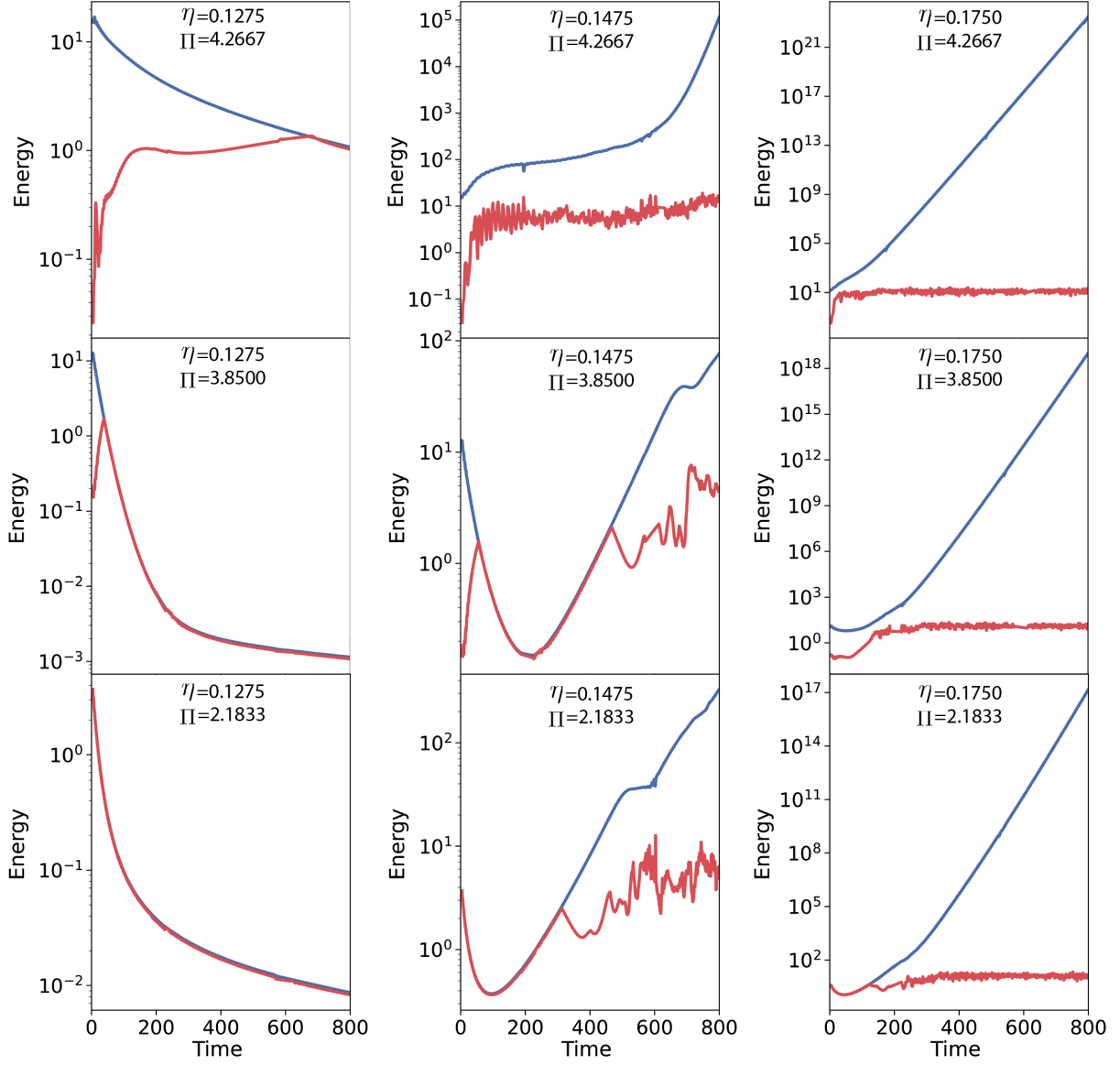


FIG. 8. The full energy $\mathcal{E}_t$ [see Eq. (C36)] (blue lines) vs the radiation energy $\mathcal{E}_r$, given by Eq. (C39) (red lines) for the numerical solution of Eq. (2) with the initial condition Eq. (3) with $\gamma = 0.1$, and the parameters η and Π shown in each plot.

only if the discrete modes are present or are created from the radiation.

Let us also mention the conserved quantities for the Ablowitz-Ladik discretization Eq. (21). The discrete analogs of the “number of particle” and the “momentum” read as

$$Q_0[\psi] = \frac{1}{a^2} \sum_{n=-\infty}^{\infty} \ln(1 + a^2 |\psi_n(t)|^2), \quad (\text{C41})$$

$$Q_1[\psi] = \sum_{n=-\infty}^{\infty} i(\psi_{n+1}^* \psi_n - \psi_n^* \psi_{n+1}). \quad (\text{C42})$$

Evaluated on the one-soliton solution Eq. (22), these quantities give

$$Q_0[\psi^{(s)}] = \frac{2\mu}{a^2}, \quad Q_1[\psi^{(s)}] = \frac{4 \sin(k) \sinh(\mu)}{a^2}. \quad (\text{C43})$$

Computing the evolution of Q_0 and Q_1 with respect to the modified equation (21), we obtain Eqs. (27) and (28), with

$$\begin{aligned} X_a^{(u)}(u, v) = & \frac{\tanh(au)}{a^3 u} \eta \sin(av) \left(\sinh(au) + \frac{au}{\cosh(au)} \right) \\ & - \frac{2\Gamma \tanh(au)}{a^3} (1 + a^2) \\ & + \frac{2\Gamma}{a^3} \sinh(au) \cos(av) + V_u^{\text{PN}}, \end{aligned} \quad (\text{C44})$$

$$X_a^{(v)}(u, v) = \frac{\sinh(au)\eta \cos(av)}{a^3 u} \left(1 - \frac{2au}{\sinh(2au)}\right) + 2 \frac{\sinh(au) \sin(av) \Gamma[1 - au \coth(au)]}{a^4 u} + V_v^{\text{PN}}. \quad (\text{C45})$$

Here the Pierls-Nabarro potentials are given by

$$V_u^{\text{PN}} = \frac{2\pi^2 \eta \sinh(au) \tanh(au) \sin(av) \cos(\frac{2\pi x_c}{a})}{a^4 u^2 \sinh(\frac{\pi^2}{au})}, \quad (\text{C46})$$

$$V_v^{\text{PN}} = \frac{2\pi^2 \sinh(au) \cos(\frac{2\pi x_c}{a}) [2\Gamma \sin(av) + a\eta \cos(av)]}{a^5 u^2 \sinh(\frac{\pi^2}{au})}. \quad (\text{C47})$$

Numerically, we convince ourselves that these terms are not relevant for the dynamics.

4. Nonlinear Schrödinger equation

For completeness let us also briefly review inverse scattering transformation for the nonlinear Schrödinger equation and outline the main steps in the derivation of Eqs. (16)–(18). Let us denote the right-hand side of Eq. (14) as $P[\psi]$. For $P[\psi] = 0$ the dynamics can be rewritten as the zero-curvature condition Eq. (C6), with matrices U and V , now given by

$$U = \begin{pmatrix} \frac{\lambda}{2i} & i\bar{\psi} \\ i\psi & -\frac{\lambda}{2i} \end{pmatrix}, \quad V = \begin{pmatrix} -i|\psi|^2 & \partial_x \bar{\psi} \\ -\partial_x \psi & i|\psi|^2 \end{pmatrix} - \lambda U. \quad (\text{C48})$$

Similar, to the IST scheme for sine-Gordon, we define the Jost solutions $T_{\pm}$ and the transfer matrix $T(\lambda)$ that connects them. It can be written as [cf. Eq. (C13)]

$$T(\lambda) = \begin{pmatrix} a(\lambda) & -\bar{b}(\lambda) \\ b(\lambda) & \bar{a}(\lambda) \end{pmatrix}. \quad (\text{C49})$$

The evolution given by Eq. (14) for $\eta = \gamma = 0$ is extremely simple on the scattering data [73]:

$$a(\lambda, t) = a(\lambda, 0), \quad b(\lambda, t) = b(\lambda, 0)e^{-i\lambda^2 t}. \quad (\text{C50})$$

This follows from the zero-curvature conditions Eq. (C6) and is also described below. The solitons correspond to the zeros of $a(\lambda)$ in the upper half-plane, $a(\lambda_i) = 0$, $\text{Im } \lambda_i > 0$. Contrary to the sine-Gordon there is only one type of soliton solutions. The associated scattering data are the proportionality coefficient between the first column of T_- and the second column of T_+ as in Eq. (C21). The evolution of this coefficient is given by

$$\mathbf{g}_j(t) = \mathbf{g}_j(0)e^{-i\lambda_j^2 t}. \quad (\text{C51})$$

The general form of the transmission coefficient $a(\lambda)$ is given by

$$a(\lambda) = \prod_{i=1}^k \frac{\lambda - \lambda_i}{\lambda - \bar{\lambda}_i} \exp\left(\frac{1}{2\pi i} \int \frac{\log(1 - |b(\mu)|^2)}{\mu - \lambda - i0} d\mu\right). \quad (\text{C52})$$

The one-soliton solution corresponds to

$$a(\lambda) = \frac{\lambda - \lambda_0}{\lambda - \lambda_0^*}, \quad b(\lambda) = 0, \quad (\text{C53})$$

for $\lambda_0 = v + iu$, $u > 0$, the corresponding profile for $\mathbf{g} = e^{x_c u + i\eta}$ is given by Eq. (15).

Now let us turn our attention to the case when the nonreciprocity and dissipation are present in the system; i.e., $P[\psi] \neq 0$. In this case the zero-curvature condition does not hold anymore; it is transformed into

$$\partial_t U - \partial_x V + [U, V] = \begin{pmatrix} 0 & -P^*[\psi] \\ P[\psi] & 0 \end{pmatrix} \equiv \mathcal{P}[\psi]. \quad (\text{C54})$$

One can show using standard ideas of the soliton perturbation theory [72,76,95] that such a change of the zero-curvature condition has the following effect on dynamics of the continuous and discrete scattering data:

$$\partial_t T(\lambda) - \frac{i\lambda^2}{2} [\sigma_z, T(\lambda)] = \int_{-\infty}^{\infty} dz T_+^{-1}(z) \mathcal{P}[\psi(z)] T_-(z), \quad (\text{C55})$$

$$\frac{d\mathbf{g}}{dt} + i\lambda_0^2 \mathbf{g} = \frac{i}{\dot{a}(\lambda_0)} \int_{-\infty}^{\infty} dz [\dot{T}_-^{(1)}(z) - \mathbf{g} \dot{T}_+^{(2)}(z)]^T \times \sigma_2 \mathcal{P}[\psi(z)] T_-^{(1)}(z), \quad (\text{C56})$$

$$\frac{d\lambda_0}{dt} = \frac{i}{\dot{a}(\lambda_0)} \int_{-\infty}^{\infty} dz [T_+^{(2)}(z)]^T \sigma_2 \mathcal{P}[\psi(z)] T_-^{(1)}(z), \quad (\text{C57})$$

here the dot means the derivative with respect to the spectral parameter λ and the right-hand side of Eqs. (C56) and (C57) should be evaluated at $\lambda = \lambda_0$. Note that the evolution Eqs. (C50) and (C51) readily follows if one neglects the right-hand side in Eqs. (C55)–(C57). To get Eqs. (16)–(18) we use the Jost functions and profile that correspond to one-solitonic profile Eq. (15).

The conserved quantities can be deduced from $a(\lambda)$ as coefficients at $\lambda \rightarrow \infty$. The first coefficient in this expansion can be related to the “total amount of matter”

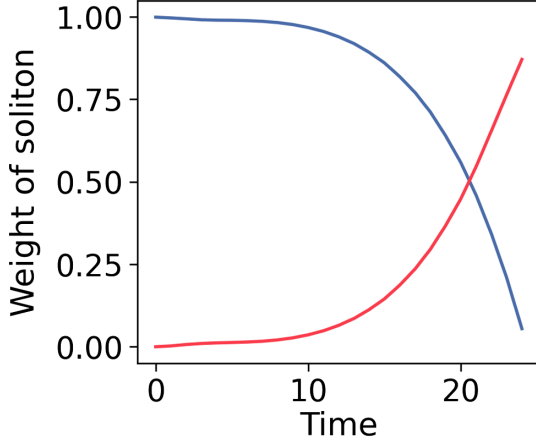


FIG. 9. The relative weight of a soliton u/I_0 (blue line) vs the radiation weight I_r/I_0 (red line). Simulation is done for $\eta = 0.133$ and $\Gamma = 0.1$, with initial conditions $u = 1.0$ and $v = 0.0$.

$I_0 = \frac{1}{2} \int_{-\infty}^{\infty} dx |\psi(x)|^2$. In the case when there is only one soliton (at $\lambda_0 = v + iu$) and radiation the value of I_0 reads

$$I_0 = u + \frac{1}{2\pi} \int_{-\infty}^{\infty} d\lambda \ln \frac{1}{|a(\lambda)|} \equiv u + I_r. \quad (\text{C58})$$

As we can independently compute I_0 , u , and I_r we can use the above relation to quantify the role of radiation in profile. We do this in Fig. 9 for the perturbed equation (14) for the initial condition $u = 1.0$ and $v = 0.0$. In this case, we see that after some time the soliton disappears completely and the whole profile starts to be dominated by the radiation. This violates presumptions that we used while deriving Eqs. (16)–(18), which explains deviations of blue lines in Fig. 4.

5. Nonlinear Schrödinger equation with constant driving

Finally, let us compare the effects of nonreciprocity with the effects of constant driving. Namely, instead of Eq. (14) we consider the following equation:

$$i\partial_t \psi + \psi_{xx} + 2|\psi|^2 \psi = -f - i\Gamma \psi + \frac{\Gamma}{2} \partial_t \psi. \quad (\text{C59})$$

Then instead of Eqs. (16)–(18) the evolution of the soliton's parameters read as

$$\frac{dv}{dt} = f \frac{\pi v \cos \Phi}{u \cosh \frac{\pi v}{2u}} - \frac{2u^2}{3} \Gamma v, \quad (\text{C60})$$

$$\frac{du}{dt} = -\pi f \frac{\cos \Phi}{\cosh \frac{\pi v}{2u}} + u\Gamma(u^2 - v^2 - 2), \quad (\text{C61})$$

$$\frac{d\Phi}{dt} = u^2 + v^2 + \frac{\pi^2 f v \sin \Phi \sinh \frac{\pi v}{2u}}{u^2 (\cosh \frac{\pi v}{2u})^2}, \quad (\text{C62})$$

$$\frac{dx_c}{dt} = 2v + \frac{\pi^2 f \sin \Phi \sinh \frac{\pi v}{2u}}{2u^2 (\cosh \frac{\pi v}{2u})^2}. \quad (\text{C63})$$

The critical point of this system can be achieved only for the extremely large value of the force $f \gg 1$ and very specific initial conditions. Indeed, the critical value of the soliton's amplitude u_c and velocity v_c are related by $u_c = \sqrt{3(2 + v_c^2)}$, and the critical value of its phase can be found from Eq. (C60) provided that the force is related to v_c as

$$f^2 = \frac{4u_c^4}{\pi^4 v_c^2} \cosh^2 \left(\frac{\pi v_c}{2u_c} \right) \left\{ v_c^2 \left[\left(\frac{\pi^2 \Gamma^2}{9} + \frac{4}{3} \right) u_c^2 + 4 \right] + (2v_c^2 + 3)^2 / \sinh^2 \left(\frac{\pi v_c}{2u_c} \right) + 9 \right\}. \quad (\text{C64})$$

This way as $v_c \rightarrow 0$, $|f| = O(1/v_c^2)$, and, as $v_c \rightarrow \infty$, $|f| = O(v_c^3)$, the minimal value is achieved for $v_c^2 \sim 1$, but even in this case the obtained force is quite large.

This is in contrast to the driving introduced by the Lugiato-Lefever equation, which would correspond to the nonlinear Schrödinger perturbation $(\gamma_R - i\gamma_I)\psi + if$ instead of the right-hand side of Eq. (14) (see Refs. [43–45]). One can easily derive a dynamical system similar to Eqs. (C60)–(C63). This system has only a static point,

$$v_c = 0, u_c = \sqrt{\gamma_R}, \sin \Phi_c = \frac{2\sqrt{\gamma_R \gamma_I}}{\pi f}, \quad (\text{C65})$$

provided that $4\gamma_R \gamma_I^2 < \pi^2 f^2$ and $\gamma_R > 0$ (see, in particular, Ref. [96]), which corresponds to a static soliton.

-
- [1] A. J. Lotka, *Analytical note on certain rhythmic relations in organic systems*, *Proc. Natl. Acad. Sci. U.S.A.* **6**, 410 (1920).
 - [2] V. Volterra, *Fluctuations in the abundance of a species considered mathematically*, *Nature (London)* **118**, 558 (1926).
 - [3] H. Sompolinsky and I. I. Kanter, *Temporal association in asymmetric neural networks*, *Phys. Rev. Lett.* **57**, 2861 (1986).
 - [4] T. Nagatani, *The physics of traffic jams*, *Rep. Prog. Phys.* **65**, 1331 (2002).
 - [5] H. Hong and S. H. Strogatz, *Kuramoto model of coupled oscillators with positive and negative coupling parameters: An example of conformist and contrarian oscillators*, *Phys. Rev. Lett.* **106**, 054102 (2011).
 - [6] N. Hatano and D. R. Nelson, *Localization transitions in non-Hermitian quantum mechanics*, *Phys. Rev. Lett.* **77**, 570 (1996).
 - [7] F. K. Kunst, E. Edvardsson, J. C. Budich, and E. J. Bergholtz, *Biorthogonal bulk-boundary correspondence*

- in non-Hermitian systems*, *Phys. Rev. Lett.* **121**, 026808 (2018).
- [8] Z. Gong, Y. Ashida, K. Kawabata, K. Takasan, S. Higashikawa, and M. Ueda, *Topological phases of non-Hermitian systems*, *Phys. Rev. X* **8**, 031079 (2018).
- [9] V. M. Martinez Alvarez, J. E. Barrios Vargas, and L. E. F. Foa Torres, *Non-Hermitian robust edge states in one dimension: Anomalous localization and eigenspace condensation at exceptional points*, *Phys. Rev. B* **97**, 121401 (2018).
- [10] S. Yao and Z. Wang, *Edge states and topological invariants of non-Hermitian systems*, *Phys. Rev. Lett.* **121**, 086803 (2018).
- [11] A. McDonald and A. A. Clerk, *Exponentially-enhanced quantum sensing with non-Hermitian lattice dynamics*, *Nat. Commun.* **11**, 5382 (2020).
- [12] S. Liu, R. Hanai, and P. B. Littlewood, *Universal scaling in one-dimensional non-reciprocal matter*, *arXiv:2503.14384*.
- [13] J. P. Mathew, J. D. Pino, and E. Verhagen, *Synthetic gauge fields for phonon transport in a nano-optomechanical system*, *Nat. Nanotechnol.* **15**, 198 (2020).
- [14] J. Del Pino, J. J. Slim, and E. Verhagen, *Non-Hermitian chiral phononics through optomechanically induced squeezing*, *Nature (London)* **606**, 82 (2022).
- [15] L. Xiao, T. Deng, K. Wang, G. Zhu, Z. Wang, W. Yi, and P. Xue, *Non-Hermitian bulk-boundary correspondence in quantum dynamics*, *Nat. Phys.* **16**, 761 (2020).
- [16] S. Weidemann, M. Kremer, T. Helbig, T. Hofmann, A. Stegmaier, M. Greiter, R. Thomale, and A. Szameit, *Topological funneling of light*, *Science* **368**, 311 (2020).
- [17] T. Liu, J.-Y. Ou, K. F. MacDonald, and N. I. Zheludev, *Photonic metamaterial analogue of a continuous time crystal*, *Nat. Phys.* **19**, 986 (2023).
- [18] E. S. Bililign, F. Balboa Usabiaga, Y. A. Ganan, A. Poncet, V. Soni, S. Magkiriadou, M. J. Shelley, D. Bartolo, and W. T. M. Irvine, *Motile dislocations knead odd crystals into whorls*, *Nat. Phys.* **18**, 212 (2021).
- [19] T. H. Tan, A. Mietke, J. Li, Y. Chen, H. Higinbotham, P. J. Foster, S. Gokhale, J. Dunkel, and N. Fakhri, *Odd dynamics of living chiral crystals*, *Nature (London)* **607**, 287 (2022).
- [20] A. Poncet and D. Bartolo, *When soft crystals defy Newton's third law: Nonreciprocal mechanics and dislocation motility*, *Phys. Rev. Lett.* **128**, 048002 (2022).
- [21] J. Colen, A. Poncet, D. Bartolo, and V. Vitelli, *Interpreting neural operators: How nonlinear waves propagate in nonreciprocal solids*, *Phys. Rev. Lett.* **133**, 107301 (2024).
- [22] S. Guillet, A. Poncet, M. Le Blay, W. T. M. Irvine, V. Vitelli, and D. Bartolo, *Melting of nonreciprocal solids: How dislocations propel and fission in flowing crystals*, *Proc. Natl. Acad. Sci. U.S.A.* **122**, e2412993122 (2025).
- [23] M. Brandenbourger, X. Locsin, E. Lerner, and C. Coulais, *Non-reciprocal robotic metamaterials*, *Nat. Commun.* **10**, 1 (2019).
- [24] M. I. N. Rosa and M. Ruzzene, *Dynamics and topology of non-Hermitian elastic lattices with non-local feedback control interactions*, *New J. Phys.* **22**, 053004 (2020).
- [25] C. Scheibner, W. T. M. Irvine, and V. Vitelli, *Non-Hermitian band topology and skin modes in active elastic media*, *Phys. Rev. Lett.* **125**, 118001 (2020).
- [26] Y. Chen, X. Li, C. Scheibner, V. Vitelli, and G. Huang, *Realization of active metamaterials with odd micropolar elasticity*, *Nat. Commun.* **12**, 5935 (2021).
- [27] W. Wang, X. Wang, and G. Ma, *Non-Hermitian morphing of topological modes*, *Nature (London)* **608**, 50 (2022).
- [28] J. Veenstra, O. Gamayun, X. Guo, A. Sarvi, C. V. Meinersen, and C. Coulais, *Non-reciprocal topological solitons in active metamaterials*, *Nature (London)* **627**, 528 (2024).
- [29] J. Veenstra, C. Scheibner, M. Brandenbourger, J. Binysh, A. Souslov, V. Vitelli, and C. Coulais, *Adaptive locomotion of active solids*, *Nature (London)* **639**, 935 (2025).
- [30] C. Coulais, R. Fleury, and J. van Wezel, *Topology and broken hermiticity*, *Nat. Phys.* **17**, 9 (2021).
- [31] E. J. Bergholtz, J. C. Budich, and F. K. Kunst, *Exceptional topology of non-Hermitian systems*, *Rev. Mod. Phys.* **93**, 015005 (2021).
- [32] S. Shankar, A. Souslov, M. J. Bowick, M. C. Marchetti, and V. Vitelli, *Topological active matter*, *Nat. Rev. Phys.* **4**, 380 (2022).
- [33] Z. You, A. Baskaran, and M. C. Marchetti, *Nonreciprocity as a generic route to traveling states*, *Proc. Natl. Acad. Sci. U.S.A.* **117**, 19767 (2020).
- [34] S. Saha, J. Agudo-Canalejo, and R. Golestanian, *Scalar active mixtures: The nonreciprocal Cahn-Hilliard model*, *Phys. Rev. X* **10**, 041009 (2020).
- [35] M. Fruchart, R. Hanai, P. B. Littlewood, and V. Vitelli, *Non-reciprocal phase transitions*, *Nature (London)* **592**, 363 (2021).
- [36] L. Braverman, C. Scheibner, B. VanSaders, and V. Vitelli, *Topological defects in solids with odd elasticity*, *Phys. Rev. Lett.* **127**, 268001 (2021).
- [37] X. M. de Wit, M. Fruchart, T. Khain, F. Toschi, and V. Vitelli, *Pattern formation by turbulent cascades*, *Nature (London)* **627**, 515 (2024).
- [38] H. Hohmann, T. Hofmann, T. Helbig, S. Imhof, H. Brand, L. K. Upreti, A. Stegmaier, A. Fritzsche, T. Müller, U. Schwingenschlögl, C. H. Lee, M. Greiter, L. W. Molenkamp, T. Kießling, and R. Thomale, *Observation of cnoidal wave localization in nonlinear topoelectric circuits*, *Phys. Rev. Res.* **5**, L012041 (2023).
- [39] B. Manda and V. Achilleos, *Insensitive edge solitons in a non-Hermitian topological lattice*, *Phys. Rev. B* **110**, L180302 (2024).
- [40] T. Dauxois and M. Peyrard, *Physics of Solitons* (Cambridge University Press, Cambridge, England, 2006).
- [41] *Dissipative Solitons: From Optics to Biology and Medicine*, Lecture Notes in Physics (Springer, Berlin, 2008).
- [42] E. A. Kuznetsov, *Solitons in a parametrically unstable plasma*, *Akad. Nauk SSSR Dokl.* **236**, 575 (1977), https://www.mathnet.ru/php/archive.phtml?wshow=paper&jrnid=dan&paperid=41246&option_lang=eng.
- [43] E. Lucas, M. Karpov, H. Guo, M. L. Gorodetsky, and T. J. Kippenberg, *Breathing dissipative solitons in optical microresonators*, *Nat. Commun.* **8**, 736 (2017).
- [44] M. Yu, J. K. Jang, Y. Okawachi, A. G. Griffith, K. Luke, S. A. Miller, X. Ji, M. Lipson, and A. L. Gaeta, *Breather soliton dynamics in microresonators*, *Nat. Commun.* **8**, 14569 (2017).
- [45] N. Pernet, P. St-Jean, D. D. Solnyshkov, G. Malpuech, N. Carlon Zambon, Q. Fontaine, B. Real, O. Jamadi, A.

- Lemaître, M. Morassi, L. Le Gratiet, T. Baptiste, A. Harouri, I. Sagnes, A. Amo, S. Ravets, and J. Bloch, *Gap solitons in a one-dimensional driven-dissipative topological lattice*, *Nat. Phys.* **18**, 678 (2022).
- [46] V. F. Nesterenko, *Propagation of nonlinear compression pulses in granular media*, *J. Appl. Mech. Tech. Phys.* **24**, 733 (1984).
- [47] C. Daraio, V. F. Nesterenko, E. B. Herbold, and S. Jin, *Tunability of solitary wave properties in one-dimensional strongly nonlinear phononic crystals*, *Phys. Rev. E* **73**, 026610 (2006).
- [48] A. Spadoni and C. Daraio, *Generation and control of sound bullets with a nonlinear acoustic lens*, *Proc. Natl. Acad. Sci. U.S.A.* **107**, 7230 (2010).
- [49] B. Deng, J. R. Raney, V. Tournat, and K. Bertoldi, *Elastic vector solitons in soft architected materials*, *Phys. Rev. Lett.* **118**, 204102 (2017).
- [50] B. Deng, P. Wang, Q. He, V. Tournat, and K. Bertoldi, *Metamaterials with amplitude gaps for elastic solitons*, *Nat. Commun.* **9**, 1 (2018).
- [51] B. Deng, V. Tournat, P. Wang, and K. Bertoldi, *Anomalous collisions of elastic vector solitons in mechanical metamaterials*, *Phys. Rev. Lett.* **122**, 044101 (2019).
- [52] B. Deng, J. Li, V. Tournat, P. K. Purohit, and K. Bertoldi, *Dynamics of mechanical metamaterials: A framework to connect phonons, nonlinear periodic waves and solitons*, *J. Mech. Phys. Solids* **147**, 104233 (2021).
- [53] B. Deng, J. R. Raney, K. Bertoldi, and V. Tournat, *Nonlinear waves in flexible mechanical metamaterials*, *J. Appl. Phys.* **130**, 040801 (2021).
- [54] D. M. Kochmann and K. Bertoldi, *Exploiting microstructural instabilities in solids and structures: From metamaterials to structural transitions*, *Appl. Mech. Rev.* **69**, 050801 (2017).
- [55] B. G. G. Chen, N. Upadhyaya, and V. Vitelli, *Nonlinear conduction via solitons in a topological mechanical insulator*, *Proc. Natl. Acad. Sci. U.S.A.* **111**, 13004 (2014).
- [56] N. Nadkarni, C. Daraio, and D. M. Kochmann, *Dynamics of periodic mechanical structures containing bistable elastic elements: From elastic to solitary wave propagation*, *Phys. Rev. E* **90**, 023204 (2014).
- [57] N. Nadkarni, A. F. Arrieta, C. Chong, D. M. Kochmann, and C. Daraio, *Unidirectional transition waves in bistable lattices*, *Phys. Rev. Lett.* **116**, 244501 (2016).
- [58] N. Nadkarni, C. Daraio, R. Abeyaratne, and D. M. Kochmann, *Universal energy transport law for dissipative and diffusive phase transitions*, *Phys. Rev. B* **93**, 104109 (2016).
- [59] J. R. Raney, N. Nadkarni, D. Daraio, D. M. Kochmann, J. A. Lewis, and K. Bertoldi, *Stable propagation of mechanical signals in soft media using stored elastic energy*, *Proc. Natl. Acad. Sci. U.S.A.* **113**, 9722 (2016).
- [60] X. Guo, M. Guzman, D. Carpentier, D. Bartolo, and C. Coullais, *Non-orientable order and non-commutative response in frustrated metamaterials*, *Nature (London)* **618**, 506 (2023).
- [61] S. Janbaz and C. Coullais, *Diffusive kinks turn kirigami into machines*, *Nat. Commun.* **15**, 1255 (2024).
- [62] P. Grelu and N. Akhmediev, *Dissipative solitons for mode-locked lasers*, *Nat. Photonics* **6**, 84 (2012).
- [63] P. Parra-Rivas, D. Gomila, M. A. Matias, S. Coen, and L. Gelens, *Dynamics of localized and patterned structures in the Lugiato-Lefever equation determine the stability and shape of optical frequency combs*, *Phys. Rev. A* **89**, 043813 (2014).
- [64] R. E. Noskov, P. A. Belov, and Y. S. Kivshar, *Subwavelength modulational instability and plasmon oscillons in nanoparticle arrays*, *Phys. Rev. Lett.* **108**, 093901 (2012).
- [65] R. Noskov, P. Belov, and Y. Kivshar, *Oscillons, solitons, and domain walls in arrays of nonlinear plasmonic nanoparticles*, *Sci. Rep.* **2**, 873 (2012).
- [66] R. E. Noskov, D. A. Smirnova, and Y. S. Kivshar, *Plasmonic kinks and walking solitons in nonlinear lattices of metal nanoparticles*, *Phil. Trans. R. Soc. A* **372**, 20140010 (2014).
- [67] Y. Hu, K. Yan, and X. Chen, *Emergent mobility edges and intermediate phases in one-dimensional quasiperiodic plasmonic chains*, *Phys. Rev. Res.* **6**, 013322 (2024).
- [68] P. B. Umbanhowar, F. Melo, and H. L. Swinney, *Localized excitations in a vertically vibrated granular layer*, *Nature (London)* **382**, 793 (1996).
- [69] H. Arbell and J. Fineberg, *Temporally harmonic oscillons in Newtonian fluids*, *Phys. Rev. Lett.* **85**, 756 (2000).
- [70] See Supplemental Material at <http://link.aps.org/supplemental/10.1103/nrv2-9h8z> for Video 1.
- [71] D. W. McLaughlin and A. C. Scott, *Perturbation analysis of fluxon dynamics*, *Phys. Rev. A* **18**, 1652 (1978).
- [72] Y. S. Kivshar and B. A. Malomed, *Dynamics of solitons in nearly integrable systems*, *Rev. Mod. Phys.* **61**, 763 (1989).
- [73] L. D. Faddeev and L. A. Takhtajan, *Hamiltonian Methods in the Theory of Solitons* (Springer, Berlin, 1987).
- [74] G. Boffetta and A. Osborne, *Computation of the direct scattering transform for the nonlinear Schroedinger equation*, *J. Comput. Phys.* **102**, 252 (1992).
- [75] A. M. Kosevich and Y. S. Kivshar, *The perturbation-induced evolution of a soliton-antisoliton pair in the sine-Gordon system*, *Fiz. Nizk. Temp.* **12**, 1270 (1982).
- [76] V. Karpman and V. Solov'ev, *A perturbation theory for soliton systems*, *Physica (Amsterdam)* **3D**, 142 (1981).
- [77] In accordance with notations of Ref. [73].
- [78] The breather velocity in the numerical solution to Eq. (2) was $\sqrt{2}/2$, which is consistent with the $\sqrt{2}$ prediction given the change of variable $t \rightarrow t/2$ done to derive Eq. (14).
- [79] M. J. Ablowitz, D. J. Kaup, A. C. Newell, and H. Segur, *The inverse scattering transform-fourier analysis for nonlinear problems*, *Stud. Appl. Math.* **53**, 249 (1974).
- [80] M. J. Ablowitz and J. F. Ladik, *Nonlinear differential-difference equations*, *J. Math. Phys. (N.Y.)* **16**, 598 (1975).
- [81] M. J. Ablowitz and J. F. Ladik, *Nonlinear differential-difference equations and Fourier analysis*, *J. Math. Phys. (N.Y.)* **17**, 1011 (1976).
- [82] Y. S. Kivshar and D. K. Campbell, *Peierls-Nabarro potential barrier for highly localized nonlinear modes*, *Phys. Rev. E* **48**, 3077 (1993).
- [83] A. A. Vakhnenko and Y. B. Gaididei, *On the motion of solitons in discrete molecular chains*, *Theor. Math. Phys.* **68**, 873 (1986).
- [84] Y. Ma, *The perturbed plane-wave solutions of the cubic Schrödinger equation*, *Stud. Appl. Math.* **60**, 43 (1979).

- [85] N. V. Alexeeva, I. V. Barashenkov, A. Dika, and R. De Sousa, *The energy-frequency diagram of the $(1 + 1)$ -dimensional Φ^4 oscillon*, *J. High Energy Phys.* **10** (2024) 136.
- [86] C.C. Wanjura, J.J. Slim, J. del Pino, M. Brunelli, E. Verhagen, and A. Nunnenkamp, *Quadrature nonreciprocity in bosonic networks without breaking time-reversal symmetry*, *Nat. Phys.* **19**, 1429 (2023).
- [87] R. Hanai, *Nonreciprocal frustration: Time crystalline order-by-disorder phenomenon and a spin-glass-like state*, *Phys. Rev. X* **14**, 011029 (2024).
- [88] T. Beatus, T. Tlusty, and R. Bar-Ziv, *Phonons in a one-dimensional microfluidic crystal*, *Nat. Phys.* **2**, 743 (2006).
- [89] M. X. Lim, A. Souslov, V. Vitelli, and H. M. Jaeger, *Cluster formation by acoustic forces and active fluctuations in levitated granular matter*, *Nat. Phys.* **15**, 460 (2019).
- [90] B. Wu, Q. Mao, B. VanSaders, and H. M. Jaeger, *Non-reciprocity and multibody interactions in acoustically levitated particle systems: A three body problem*, *arXiv: 2503.12327* [Phys. Rev. E (to be published)].
- [91] M. Reisenbauer, H. Rudolph, L. Egyed, K. Hornberger, A. V. Zasedatelev, M. Abuzarli, B. A. Stickler, and U. Delić, *Non-Hermitian dynamics and non-reciprocity of optically coupled nanoparticles*, *Nat. Phys.* **20**, 1629 (2024).
- [92] Non-reciprocal breathing solitons, <https://uva-hva.gitlab.host/published-projects/non-reciprocal-breathing-solitons.git>.
- [93] D. Zwicker, *py-pde: A PYTHON package for solving partial differential equations*, *J. Open Source Software* **5**, 2158 (2020).
- [94] O. Gamayun and M. Semenyakin, *Soliton splitting in quenched classical integrable systems*, *J. Phys. A* **49**, 335201 (2016).
- [95] D. J. Kaup and A. C. Newell, *Solitons as particles, oscillators, and in slowly changing media: A singular perturbation theory*, *Proc. R. Soc. A* **361**, 413 (1978).
- [96] T. Herr, M. L. Gorodetsky, and T. J. Kippenberg, *Dissipative Kerr solitons in optical microresonators*, *Nonlinear Optical Cavity Dynamics* (Wiley, New York, 2015).